

Why Do Workers Dislike Inflation?

Wage Erosion and Conflict Costs*

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Abstract

How costly is inflation to workers? Answers to this question have focused on the path of real wages during inflationary periods. We argue that workers must take costly actions (“conflict”) to have nominal wages catch up with inflation, meaning there are welfare costs even if real wages do not fall as inflation rises. We study a menu-cost style model, where workers choose whether to engage in conflict with employers to secure a wage increase. We show that, following a rise in inflation, wage catch-up resulting from more frequent conflict does not raise welfare. Instead, the impact of inflation on worker welfare is determined by what we call “wage erosion”—how inflation would lower real wages if workers’ conflict decisions did not respond to inflation. As a result, using observed wage growth to measure worker welfare understates the costs of inflation. We conduct a survey showing that workers are willing to sacrifice 1.75% of their wages to avoid conflict. Calibrating the model to survey data, incorporating conflict significantly raises the costs of inflation for workers.

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1 Introduction

People think inflation is one of the United States’ worst problems ([Pew Research Center, 2022, 2023](#)). Why do people dislike inflation so much? One reason could be that prices rise faster than nominal wages when inflation is high, meaning real wages fall and workers become poorer ([Shiller, 1997](#); [Stantcheva, 2024](#); [Afrouzi, Dietrich, Myrseth, Priftis, and Schoenle, 2024b](#)). A classic view, for instance from [Fischer and Modigliani \(1978\)](#) or [Mankiw’s \(2020\)](#) textbook, suggests that this cost of inflation is small. The argument is that nominal wages generally keep up with prices after an inflationary shock. As a result, real wages do not persistently fall and workers do not suffer much.

This paper argues that inflation imposes costs on workers beyond its impact on real wages. We start from the observation that employers do not automatically give workers raises when inflation is high. Instead, workers have to fight for these raises, which places them in conflict with employers. We propose a standard and tractable menu-cost style model that incorporates the role that costly conflict plays in determining wage growth. We show that accounting for “conflict costs” meaningfully changes our understanding of the costs of inflation, both analytically and quantitatively. In this setting, what matters for workers’ welfare is not how inflation impacts real wages, but rather how inflation would affect real wages if workers did not choose to engage in more conflict as inflation rises, a concept we term “wage erosion.” Our framework delivers a direct mapping between conflict costs and the wages that workers would sacrifice to avoid conflict, which we measure from a survey of US workers to be 1.75% of their wages. Combining our model with these estimates, we find that conflict more than doubles the costs of inflation to workers relative to the costs of inflation implied by falling real wages alone. We conclude that a “conflict cost” model is a tractable and quantitatively relevant way of understanding the welfare costs of inflation for workers.¹

We start the paper with motivating survey evidence about the relationship between conflict and inflation. We fielded a survey to 3000 US workers at the start of 2024, in the aftermath of the post pandemic inflation, and arrive at two conclusions. First, we find that conflict is important for determining wage growth. A significant portion of workers say they took costly actions—that is, they engaged in conflict—to achieve higher wage growth than their employer offered. These actions include having tough conversations with employers about pay, partaking in union activity, or soliciting job offers. We

¹By the same logic, “conflict costs” can also be relevant for the welfare costs of other shocks that require nominal wage adjustments.

find that these costly actions lead to higher wages, as participants who took these actions report that their wage growth in 2023 was 3 percentage points higher than what their employers initially offered. Conversely, those who did not take the costly actions believe conflict would have raised wages by 2 percentage points, suggesting sizable conflict costs that offset the benefit of higher wages.

Second, we find that conflict rises with inflation. Respondents say that the costly actions were primarily motivated by wanting wages to keep up with inflation. Additionally, when asked how they would behave at different rates of inflation, respondents were more likely to engage in conflict with employers when inflation was higher. We complement this result with observational evidence that conflict between workers and firms is more likely when inflation is higher. In cross-country panel regressions using data from 1964 to 2022, we document a robust positive correlation between inflation and conflict, proxied by labor market strikes.

We propose a tractable "conflict cost" model to capture this state-dependent nature of wage setting, and deliver analytical and quantitative insights about how conflict affects the welfare costs of inflation. Motivated by our survey evidence, workers in the model receive a default nominal wage offer from their employer. Unless the offer is fully indexed to inflation, the worker's offered real wage falls when inflation rises. In response, workers optimally choose whether to engage in conflict with employers. Conflict increases the worker's nominal wage beyond their employer's offer, ensuring that it keeps up with inflation. However, conflict is costly to workers. Similar to the menu cost literature, workers face idiosyncratic productivity shocks, and the "conflict cost" of increasing wages beyond the employer's offer takes the "Calvo-plus" form of [Nakamura and Steinsson \(2010\)](#) and [Auclert, Rigato, Rognlie, and Straub \(2024a\)](#). In our model, costly conflict is more likely when inflation increases, consistent with the state-dependence of our motivating evidence. Intuitively, as inflation rises, without conflict, workers' real wages fall. Thus, the wage gain they can achieve through conflict grows, leading more workers to choose to take costly actions.

Our main analytical result characterizes the welfare costs of inflation shocks to workers. The path of real wages is no longer sufficient to inform worker welfare in this setting. In particular, wage catch-up after inflation which workers achieve through more frequent conflict does not raise welfare. On the margin, the extra conflict costs paid by workers to ensure higher wages cancels out the benefits of the higher wages. The cancellation follows from worker optimality and the envelope theorem of [Milgrom and Segal \(2002\)](#), applied to discrete conflict choices. Instead, the first-order impact of inflation shocks on worker welfare is determined by "wage erosion", which we define as how inflation

shocks would affect real wages if workers' conflict decisions had not moved with inflation.² As such, the welfare costs of inflation in the labor market can be significant even if real wages do not fall, as workers must take more frequent costly actions to ensure wage catch-up. Unlike falls in the real wage, which redistribute from workers to firms, conflict costs potentially create aggregate losses too.

How quantitatively important is conflict for the welfare costs of inflation? We answer this question by calibrating our model using additional, tailored survey questions. We then calculate the welfare costs of inflationary shocks to workers within that environment. We designed survey questions to directly inform the two parameters governing the importance of conflict in the model, namely: 1) the cost to an individual worker of conflict with an employer, and 2) the extent to which employers' default wage offers are indexed to inflation. In the survey, we find that conflict with employers is costly to workers – the median worker would sacrifice 1.75 percent of their wage to avoid conflict. We validate the survey estimate by showing that our measure of conflict costs predicts workers' reported conflict decisions in 2023. Second, workers believe that employers' wage offers are weakly indexed to inflation. Specifically, we asked workers to consider various hypothetical levels of inflation. Absent conflict, workers believe that employers would raise their wage offer by 0.05 percentage points for every percentage point increase in inflation.³

Our analysis shows that, when calibrated to match these survey moments, conflict significantly raises the welfare costs of inflation for workers. To solve the model, we rely on recent advances in Sequence-Space Jacobian methods in [Auclert, Bardóczy, Rognlie, and Straub \(2021\)](#) and [Auclert et al. \(2024a\)](#). In response to either transitory or persistent inflation shocks, incorporating conflict more than doubles the overall costs of inflation to workers. In various extensions, conflict continues to significantly increase the costs of inflation—for instance, at significantly higher levels of default wage

²Our definition of wage erosion relates to the intensive margins of price adjustment in [Auclert et al. \(2024a\)](#), but we study wage instead of price setting and normative rather than positive implications.

³Our approach of directly measuring conflict costs differs from the menu-cost literature, which infers menu-costs from the moments of the price change distribution (e.g., [Alvarez, Le Bihan, and Lippi, 2016](#)). However, different from the menu-cost model, in our conflict-cost model, wages change without conflict for many reasons. For instance, the default wage offered by employers can grow due to partial indexation to inflation shocks. As a result, we would have to place hard-to-verify assumptions on the structure of the model to differentiate wage changes that arise with conflict from those that arise without conflict and to infer conflict costs from the distribution of wage changes. Our approach instead provides direct evidence on how much workers dislike conflict.

indexation than our baseline calibration; when wages and employment are determined in general equilibrium, allowing inflation to “grease the wheels” of the labor market (Blanco and Drenik, 2023); or when studying the inflation of 2021-23. In sum, the “conflict cost” model is a quantitatively relevant and tractable way of understanding the welfare costs of inflation for workers.

Beyond the specific application to the costs of inflation, our conflict cost model is a natural way of introducing state-dependent wage setting into New Keynesian models. To model sluggish wage adjustments, the New Keynesian literature typically assumes time-dependent wage setting—for instance Erceg, Henderson, and Levin (2000), Gertler, Sala, and Trigari (2008), Gertler and Trigari (2009), Galí (2011), Galí, Smets, and Wouters (2012), Gertler, Huckfeldt, and Trigari (2020), Chodorow-Reich et al. (2023), Auclert, Bardóczy, and Rognlie (2023) and Auclert, Rognlie, and Straub (2024b). State-dependent wage setting is a natural alternative. After all, the state-dependent approach is common when modeling price setting (e.g., Gertler and Leahy, 2008), and state dependence is consistent with empirical evidence in our survey. Moreover there are distinct positive and normative implications compared to the time-dependent approach.⁴

Related literature. This paper contributes to the large literature on the costs of inflation. Previous work identifies inflation costs from a range of mechanisms, such as “shoe leather costs” of holding less money (e.g., Bailey, 1956; Friedman, 1969; İmrohoroglu, 1992; Lucas, 2000); “menu costs” from changing prices and the associated price distortions (e.g. Burstein and Hellwig, 2008; Nakamura, Steinsson, Sun, and Villar, 2018; Alvarez, Beraja, Gonzalez-Rozada, and Neumeyer, 2019; Caratelli and Halperin, 2023); tax distortions (e.g. Feldstein, Green, and Sheshinski, 1978; Altig, Auerbach, Eidschun, Korklikoff, and Ye, Forthcoming); uncertainty due to volatile inflation (Friedman, 1977); cognitive costs due to complexity and difficulty in budgeting (Shiller 1997; Binetti, Nuzzi, and Stantcheva 2024); and broader social and economic costs such as declining trust in government. Besides these other costs, we argue for significant “conflict costs” of inflation via the labor market.

A range of papers study how inflation leads to welfare costs in the labor market by lowering real wages. Surveys from Shiller (1997), Stantcheva (2024) and Afrouzi et al. (2024b) show that people dislike inflation in large part because they believe high inflation lowers real wages. Our mechanism suggests a reason for this view. People know that if prices have risen faster than the default nominal wage offered by their employer, they must engage in painful conflict with their employer to rectify the

⁴Previous work by Takahashi (2018), Jo (2019), Costain, Nakov, and Petit (2019) and Blanco and Drenik (2023) studies positive implications of state dependence in wage setting.

situation. [Del Canto, Grigsby, Qian, and Walsh \(2023\)](#) operationalize a general sufficient-statistic approach in order to estimate the effect of inflationary shocks on welfare, taking into account, amongst other channels, the effect of inflation on real wages.⁵ We show that the behavior of real wages is an important but incomplete account of the costs of inflation that operate in the labor market. Rather, having nominal wages keep up with prices entails significant additional welfare costs due to conflict.

In arguing that inflation leads workers to take costly actions, our paper relates to some previous evidence. [Hajdini, Knotek, Leer, Pedemonte, Rich, and Schoenle \(2023\)](#) implement a large randomized controlled trial in which certain workers receive an information treatment that raises their inflation expectations. Workers believe that higher inflation causes lower real wages, leading many to search for a new job with higher wages. [Pilossoph and Ryngaert \(2024\)](#) and [Pilossoph, Ryngaert, and Wedewer \(2024\)](#) show with observational data and survey hypotheticals that workers with higher inflation expectations are more likely to search for new jobs in order to secure nominal pay increases. [Stantcheva \(2024\)](#) surveys workers, and finds that they react to inflation by taking costly actions such as searching for other jobs or asking for pay increases. Besides providing additional survey evidence consistent with these papers, we model the welfare effects of inflation due to these costly actions, and quantify the costs with our survey and model.

In contemporaneous work, [Afrouzi, Blanco, Drenik, and Hurst \(2024a\)](#) explain how job search affects wage dynamics and welfare during inflationary episodes. They implement a state-of-the-art quantitative labor-search model with nominal rigidity, disciplined with information on labor market flows. We instead model a wide range of costly actions via a reduced form “conflict cost,” within a menu-cost style model disciplined by survey evidence. Despite these differences, the two quite different approaches reach a similar conclusion: costly actions to secure wage increases are a key feature of inflationary episodes.

There is an empirical literature studying whether nominal wages keep up with inflation. An older literature found that aggregate nominal wages tended to keep up with inflation ([Kessel and Alchian, 1960](#); [Bach and Stephenson, 1974](#)). Modern work such [Blanco, Drenik, and Zaratiegui \(2024\)](#) em-

⁵[Ferreira, Leiva, Nuño, Ortiz, Rodrigo, and Vazquez \(2023\)](#) and [Pallotti, Paz-Pardo, Slacalek, Tristani, and Violante \(2023\)](#) apply a similar sufficient statistic approach. [Auclert \(2019\)](#) develops the sufficient-statistic approach, in order to analyze the positive implications of monetary policy shocks. [Doepke and Schneider \(2006\)](#) also estimate the redistributive effect of inflation via asset markets, as opposed to labor markets.

phasizes that wage dynamics during inflationary episodes depend on factors such as the nature of the inflationary shock and especially workers' position in the wage distribution. Consistent with this view, poorer workers experienced stronger real wage growth during the post pandemic inflation (Autor, Dube, and McGrew, 2023). A common finding is that wages at least partly keep up with inflation for most workers, motivating us to study the associated costs.

Lorenzoni and Werning (2023a,b) study related themes about inflation and conflict. The aspect of conflict studied in these papers is disagreement between workers and firms over relative prices. In this way, conflict is a proximate cause of inflation dynamics. We study a related but different aspect of conflict: how inflation makes workers seek conflict with their employers to raise wages. Rather than investigate the cause of inflation, we ask how conflict affects the costs of inflation to workers.

Outline. Section 2 discusses the design of the survey. Section 3 presents motivating evidence about wages, inflation and conflict. Section 4 contains our conflict-cost model and main analytical result. Section 5 describes how we elicit information about conflict costs. Section 6 quantifies how conflict affects the cost of inflation, using the model and our measure of conflict costs. The Supplementary Appendix and Additional Materials contains proofs, additional analyses, and the survey questionnaire.

2 Survey Design

We developed an original survey to achieve two goals. First, the survey provides evidence about the relationship between inflation and conflict to inform the structure of our model. Second, we designed questions tailored to calibrate the key parameters of the model and quantify how conflict affects the costs of inflation.

We ran the survey between February and March 2024. We used Prolific, a survey marketplace that recruits respondents for research studies. We collected a total of 3,000 responses, with participation limited to individuals aged 22 to 60, employed either full-time or part-time, and not self-employed. We used an attention check to filter out careless participants.⁶ We further imposed quotas, requiring a certain number of respondents within groups by gender, education and political affiliation. The quotas for gender and education targeted population shares from Current Population Survey (CPS) data

⁶Participants who failed the attention check were compensated for their participation and asked to return their submissions, allowing other respondents to take their place.

for March 2023. The quota for political affiliation targeted data from Gallup 2024. Participants took an average of 7 minutes and 15 seconds to complete our survey and were rewarded for completing the survey at an average rate of \$1.40, equivalent to \$12 per hour.

Appendix Table B.1 compares our sample characteristics to the US population. Our sample broadly matches the demographic distribution of the US population, albeit with a higher representation of individuals in their thirties and a smaller proportion of respondents in their fifties. Additionally, our sample includes a lower share of white individuals and a higher share of mixed race individuals.

The survey consisted mostly of closed-ended questions. However, following best practices for “hybrid” open- and close-ended questions, we include an “Other” option in several questions (Stantcheva, 2023). This option allowed participants to express their thoughts in an open-ended fashion, enabling us to avoid imposing our preconceptions. In the survey, we asked questions about “pay growth” and not “wage growth”. In preliminary tests, we discovered that survey respondents found the “pay growth” language easier to understand. However, for consistency with the rest of the paper, we refer to “wage growth” as we describe our results. The full questionnaire is in Appendix D.

3 Motivating Evidence: Wages, Inflation and Conflict

This section presents four pieces of motivating evidence relating conflict, wage growth, and inflation. Our first three findings establish the importance of conflict for wage setting in 2023. Our fourth finding is that conflict rises with inflation. This evidence motivates the structure of our model. In Section 5, we will further discuss the tailored questions in the second part of the survey that we will use to identify the key parameters of the model.

Finding 1: Workers choose between accepting their employer’s “default wage” and engaging in conflict. Our survey shows that the employer’s default wage offer is key for determining wage growth. After eliciting the respondent’s wage growth in 2023, we asked respondents whether they accepted the wage offered to them by their employers, or whether they took costly actions in order to secure this wage growth. 79% of workers accepted the wage offer made by their employer, while 21% of workers took actions to secure their wage growth (see Appendix Figure B.1). Appendix Figures B.2 and B.3 shows how the prevalence of conflict varies with observable characteristics of the worker. We find modest heterogeneity, with those who are younger, have higher incomes, or work in the government being slightly more likely to accept the default wage offers. The only group that was much less likely

Figure 1: The Effectiveness of Conflict



Note: Panel A and B depict the distribution of reported wage growth during 2023 and the hypothetical wage growth respondents reported they would have received if no actions had been taken or if actions had been taken to achieve a higher pay, respectively. The medians of both distributions are highlighted in each subfigure. The data range has been truncated, with values ranging from a minimum of -5% to a maximum of 15%. Panel A restricts to respondents who took actions to achieve a higher pay during 2023, asking the question “Above, you indicated that you got a pay raise by either initiating a difficult conversation with your employer about your pay, searching for a higher paying job with other employers or switching employers in order to get a raise.” Panel B restricts to respondents who accepted their employers’ default wage during 2023, asking the question “[w]hat pay growth do you think you could have attained this past year if you had taken actions such as initiating a difficult conversation with your employer to ask for a raise, searching for higher paying jobs with other employers, or switching employers in order to get a raise?”.

to have accepted the default wage offer were those in unionized sectors.

We then investigated which actions respondents took in order to increase their pay. Workers took a diverse set of actions, such as having difficult conversations with their employer, securing an offer from another employer to raise pay with their current employer, or having a union negotiate on their behalf (see Appendix Figure B.4). This wide range of actions will motivate us to model conflict as a reduced form cost paid by the worker to secure a pay rise.

Finding 2: Conflict raised wages. Workers who engaged in conflict believe these actions increase their wage growth. In orange in the left panel of Figure 1, we plot the wage growth that these action takers reported having received in 2023. We also asked respondents what wage growth they believe they would have received without taking actions, and plot this wage growth in blue. The distribution of hypothetical wage growth without actions is generally to the left of actual wage growth, with a median wage growth of 0 percentage points, compared to a median wage growth with these actions of 3 percentage points.⁷

There is a similar pattern, both quantitatively and qualitatively, from directly comparing the wage

⁷The left panel of Appendix Figure B.5 plots the difference in wage growth, with or without action, for each worker who takes actions to increase their pay. The average worker reported that actions raised their wage by 2%.

growth for those workers in our survey who engaged in conflict in 2023 to those who did not. Those who engaged in conflict experienced wage growth of 5.1% in 2023, compared to 3.1% for those who accepted their employer's offer.

Finding 3: Workers who do not engage in conflict believed it would have raised their wages.

What about the workers who did not engage in conflict in 2023? One possibility is that workers did not engage in conflict because they believed it would not raise wages. Another possibility is that conflict could have raised their wages, but these workers disliked taking such actions. The evidence suggests the latter. In the right panel of Figure 1, we plot the wage growth of people who did not take actions, as well as the wage growth they believe they would have received if they had taken actions. These workers believe that median pay would have been 2 percentage points higher if they had taken actions.⁸ Evidently, these workers perceive substantial costs associated with conflict, as they were willing to sacrifice significant wage growth to avoid taking action.⁹

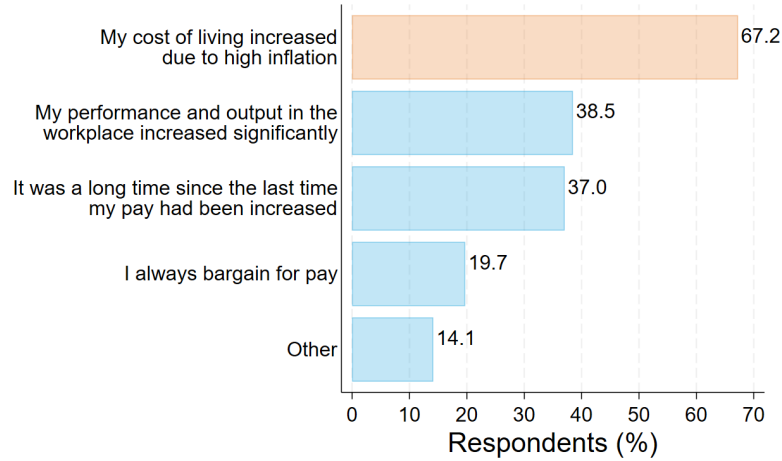
Finding 4: Inflation leads to conflict. Our final finding considers when workers choose to engage in conflict. We present several pieces of evidence that suggest that inflation leads to conflict. First, we asked workers taking costly actions in 2023 to report why they chose to act.¹⁰ The answers, in Figure 2, show that rising inflation was their main motivation, with 67% of the action takers reporting that they needed to combat a high cost of living. The next most important reason, that people deserved higher pay due to their performance, mattered for only 38% of respondents. Reassuringly, only 14% of respondents selected the “other” option.

⁸The right panel of Appendix Figure B.5 plots the difference in wage growth, with or without action, for each worker who did not take actions to increase their pay. The average worker who did not take actions believes they sacrificed 1.8 percent of wages by accepting the employer's offer.

⁹The qualitative responses explaining why workers chose to accept their employer's default wage offer also suggest that conflict is costly (see Appendix Figure B.6).. The most common reason to accept the offer is a lack of alternative job options, suggesting high costs to searching for another job in order to raise wages at the current job. The second most common reason is that negotiations are not allowed by the company. Overcoming a norm that wages cannot be negotiated is difficult, implying high conflict costs for this group.

¹⁰To do so, we followed a procedure to avoid imposing preconceptions. We asked our pilot of 100 respondents to discuss in open-ended form why they took costly actions, and grouped their reasons into a set of categories that we presented to the full survey. Again, we allow respondents to select an “other” option, and randomized the order in which the categories were presented.

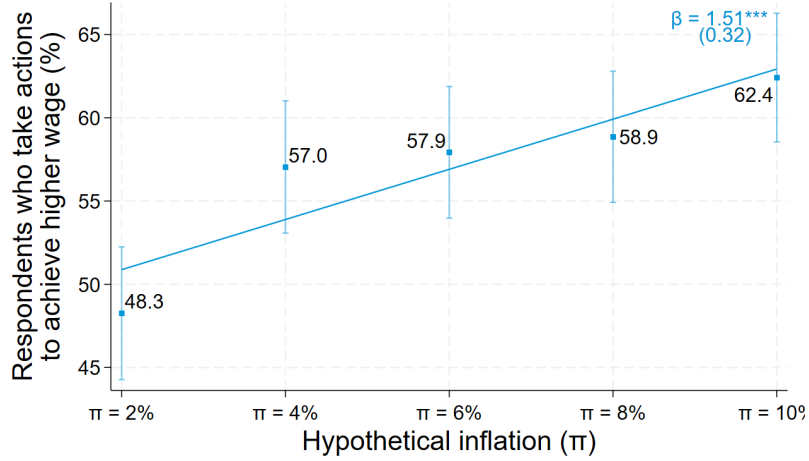
Figure 2: Motivation for Costly Actions in 2023



Note: The figure shows the percentage of survey participants who stated their motivations to take costly actions to achieve a higher pay during 2023, answering the question "[w]hat was your, or your union's, motivation for taking actions in order to secure a pay increase in 2023?" Each bar in the figure represents the following answer choices in order: "My cost of living increased due to high inflation, therefore I needed more money to fund my spending and saving plans"; "My performance and output in the workplace increased significantly"; "It was a long time since the last time my pay had been increased"; "I always bargain for pay"; and "Other, please add additional comments below". The data includes only respondents who indicated that they, either individually or through their unions, took actions to achieve higher pay during 2023.

Second, we used a hypothetical question to test whether workers are more likely to engage in conflict when inflation is high. We randomly assigned participants into five equally sized groups, each of which were offered a hypothetical scenario in which inflation was expected to be 2%, 4%, 6%, 8% or 10% over the next 12 months. We stipulated that other aspects of employment such as hours worked and their employer would remain the same. We first asked respondents what wage they thought their employer would offer them in that scenario. We then asked whether respondents would choose to take costly actions to achieve higher wages. Figure 3 shows the results. The y-axis shows the fraction of respondents who, when given a particular hypothetical scenario, say they would engage in conflict with their employer to achieve higher wages. The x-axis is hypothetical inflation under each scenario. When hypothetical inflation is 2%, 48.3% of respondents say they would take costly actions to achieve higher wage growth. However, when inflation is hypothetically 10%, more than 60% of respondents would take action. The regression line indicates that for every percentage point increase in inflation, people believe they would be 1.5 percentage points more likely to take actions that put them in conflict with their employer. One can usefully compare this result to [Hajdini et al. \(2023\)](#), which exogenously raises certain workers' inflation expectations using information treatments within a randomized controlled trial. The paper finds that inflation causes workers to search more, which is

Figure 3: Inflation and the Probability of Conflict



Note: This scatterplot displays the relationship between the indicator of whether respondents would take actions to secure a wage higher than their employer's default offer under a hypothetical inflation scenario, and the hypothetical inflation rate. The indicator is equal to one if respondents would take actions to secure a higher wage; otherwise zero. Standard errors are in brackets. The stars indicate levels of statistical significance: 1% (***), 5% (**), and 10% (*). The sample is all respondents. Respondents answered the following question. "Consider a hypothetical situation in which inflation is expected to be $x\%$ in the next 12 months. Suppose that you are working at the same job at the same place you currently work, and working the same number of hours. Would you accept your employer's offer without taking any actions to increase your pay or would you do your best to increase your pay using any strategies at your disposal?" In the question, x was 2%, 4%, 6%, 8% or 10%.

a specific kind of costly action. [Pilossoph and Ryngaert \(2024\)](#) and [Pilossoph, Ryngaert, and Wedewer \(2024\)](#) also document that workers with higher inflation expectations are more likely to search for new jobs. Our hypothetical question arrives at a similar conclusion.

Third, we find similar patterns in observational data. Systematic data on conflict between workers and firms is typically hard to collect. One exception is unions, who publicly report when they choose to conflict with employers by going on strike. Consistent with our survey result, strikes rise with inflation. Specifically, we estimate

$$\Delta \log(\text{Strike}_{c,t}) = \beta \Delta \pi_{c,t} + \gamma_c + \gamma_t + \epsilon_{c,t} \quad (1)$$

where $\text{Strike}_{c,t}$ is the number of workers on strike in country c in year t , sourced from the International Labour Organization, and $\Delta \pi_{c,t}$ is the 5-year change in the rate of inflation in country c . γ_c and γ_t are country and year fixed effects, respectively. The combination of differencing and including fixed effects removes country specific trends in strikes and inflation. We include data on 78 countries from 1974-2022. We are interested in β , which captures the relationship between the 5-year growth rate of

the number of workers on strike in a given country and the change in inflation in that country over the corresponding 5-year period. Appendix Figure B.7 shows the binned scatterplot underlying the estimate of β in Equation (1), after controlling for fixed effects. There is a clear positive relationship – when inflation in a country rises by 10 percentage points over a 5-year period, the number of workers on strike increases by 29 percent. Appendix Table B.2 shows that this cross-sectional correlation is robust to various choices such as the time difference, measure of inflation or sample period. This relationship is consistent with the premise that increases in inflation leads to more conflict between workers and firms. However, we caution that other reasons for the relationship are also possible—for instance, higher labor market power could cause more industrial action and, as a result, more inflation.

4 A Conflict-Cost Model

Motivated by the evidence in the previous section, we develop a “conflict-cost model” to investigate how conflict affects the welfare costs of inflation to workers. In the model, which focuses on the worker’s decision problem, workers receive a nominal default wage offer from their employer that may not be fully indexed to inflation. Workers optimally choose whether to engage in conflict with their employer to have wages catch up with inflation. We show that, following an inflation shock, wage catch-up resulting from more frequent conflict does not raise welfare to first order. The impact of the inflation shock on workers’ welfare is determined solely by wage erosion—how the inflation shock would lower real wages if workers’ conflict decisions did not respond to inflation. As such, the behavior of real wages misses welfare costs due to more frequent conflict.

4.1 The Worker’s Problem

Time is discrete and indexed by $t \in \{0, 1, \dots\}$. The economy is populated by a continuum of workers $i \in [0, 1]$. Each worker’s preference is given by

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t (\log c_{i,t} - \kappa_{i,t} \mathcal{I}_{i,t}) \right], \quad (2)$$

where $c_{i,t}$ is worker's consumption, over which they have logarithmic utility.¹¹ $\mathcal{I}_{i,t}$ is an indicator function, which takes a value of one if the worker chooses to take costly actions that place them in conflict with their employer in order to increase pay. $\kappa_{i,t}$ is the “conflict cost”, i.e., the utility cost to worker i of taking the costly action at time t . The conflict cost takes the “Calvo-plus” form of Nakamura and Steinsson (2010) and Auclert et al. (2024a).¹² Specifically, with probability $1 - \lambda$, the worker incurs a utility cost $\kappa > 0$ to increase pay, meaning $\kappa_{i,t} = \kappa$. With probability λ , conflict is costless to the worker, meaning $\kappa_{i,t} = 0$. The cost $\kappa_{i,t}$ is i.i.d. over time and across workers.

Each worker i receives a real wage $w_{i,t} = W_{i,t}/P_t$, where $W_{i,t}$ is the nominal wage and P_t is the price level. If the worker does not take actions to increase pay ($\mathcal{I}_{i,t} = 0$), they earn a default real wage $w_{i,t} = w_{i,t}^d = w_{i,t-1} e^{\alpha - \pi^{ss} - (1-\gamma)(\pi_t - \pi^{ss})}$, which is $W_{i,t}^d = W_{i,t-1} e^{\alpha + \gamma(\pi_t - \pi^{ss})}$ in nominal terms. Here, α denotes the growth rate of the default nominal wage at steady state inflation π^{ss} , $\gamma \in [0, 1]$ is the degree of indexation to inflation shocks ($\gamma = 0$ is no indexation and $\gamma = 1$ is full indexation), and $\pi_t = \log(P_t/P_{t-1})$ is the inflation rate. If the worker takes actions to increase their pay ($\mathcal{I}_{i,t} = 1$), they can raise it to a conflict-induced (real) wage $w_{i,t} = w_{i,t}^*$ that keeps up with inflation and productivity.

The conflict-induced real wage $w_{i,t}^*$ is exogenous and grows in line with productivity, following

$$\log w_{i,t}^* = \log w_{i,t-1}^* + g + z_{i,t}, \quad (3)$$

where $z_{i,t}$ represents idiosyncratic productivity shocks and g represents trend productivity growth. The idiosyncratic shock has a mean of $\mathbb{E}[z_{i,t}] = 0$, is i.i.d. across workers and time, and is independent of $\kappa_{i,t}$. This implies that the conflict-induced wage is fully indexed to inflation. In Section 4.3, we also demonstrate how our main result extends to cases where conflict-induced nominal wages are partially indexed or overshoot in response to inflation shocks.

In sum, workers receive wage increases without costly conflict in two ways. First, workers receive wage growth via the firm's default wage offer, governed by parameters α and γ . Second, a fraction λ of workers receive a “free” catch-up opportunity in each period, which ensures wages keep up with inflation and productivity growth. Such a free wage increase might come from workers having a low cost of conflict for idiosyncratic reasons. It may also arise from firms unsolicitedly offering workers

¹¹The log utility case provides a clean benchmark because, in this case, conflict decisions are independent of the level of wages that a worker has.

¹²We adopt “Calvo-plus” costs for simplicity, however our main results hold with more a general distribution of costs as in Alvarez, Lippi, and Oskolkov (2022). See Section 4.3 for details.

wage increases to ensure their wages keep up with inflation and productivity growth, thereby avoiding conflict.

In the main analysis, we study the case where the worker is hand-to-mouth and $c_{i,t} = w_{i,t}$. In extensions below, we also study the case in which the worker faces a standard borrowing constraint and verify that our main conclusion stands.

To summarize the worker's problem conveniently, we introduce a "wage gap", defined as the difference between the actual wage and the conflict-induced wage, $x_{i,t} \equiv \log w_{i,t} - \log w_{i,t}^*$. Based on the process for real wages, the dynamics of the wage gap are given by

$$x_{i,t} = \begin{cases} x_{i,t-1} - (\mu + z_{i,t}) - (1 - \gamma)(\pi_t - \pi^{ss}) & \text{if } \mathcal{J}_{i,t} = 0 \\ 0 & \text{if } \mathcal{J}_{i,t} = 1 \end{cases}, \quad (4)$$

where $\mu \equiv g - \alpha + \pi^{ss} \geq 0$ parametrizes the drift of the wage gap in steady state.

We make some auxiliary assumptions about the distribution of the idiosyncratic shock: $z_{i,t}$ is continuously distributed over a support $[\underline{z}, \infty)$, where $\mu + \underline{z} \geq 0$. We use $f(z_{i,t})$ to denote its probability density function, which has a mean of $\mathbb{E}[z_{i,t}] = 0$. The continuity of the distribution of $z_{i,t}$ guarantees that the steady-state distribution of wage gaps has no mass point, which facilitates our analytical results. The lower bound $\underline{z}$ in the support of $z_{i,t}$ guarantees that the steady-state distribution of wage gaps has a non-positive support, so, for all i and t , $w_{i,t} \leq w_{i,t}^*$ at steady-state inflation. That is, the worker's productivity shock realization is never so negative that their default wage $w_{i,t}^d$ is higher than the conflict-induced wage $w_{i,t}^*$.

This setup captures several features of wage setting found in Section 2. Workers choose between accepting a default wage offered to them by the employer or increasing pay by taking costly actions that place them in conflict with their employers. The specific actions that lead to this pay growth could be diverse and depend on the specifics of the job (e.g., initiating a difficult conversation with the employer, searching for and switching to higher-paying jobs, or partaking in industrial action). We model their costs with a single reduced form parameter κ , representing time, monetary, and psychological costs. The actions ensure a wage $w_{i,t}^*$ that keeps up with both inflation and productivity growth. In the absence of these actions, workers receive a default wage offered by the employer that may not be fully indexed to inflation shocks ($\gamma < 1$). In other words, the default contract between workers and employers is potentially incomplete with respect to inflation shocks (Grossman and Hart, 1986; Hart

and Moore, 1990), with the degree of incompleteness indexed by a parameter γ .

Alternate more sophisticated wage setting policies could feature firms setting wages to prevent conflict entirely, in order to avoid paying higher conflict-induced wages ($w_{i,t}^*$) after worker conflict. For instance, firms could always offer wages that are just high enough that workers choose not to engage in conflict, in essence marking wages down by exactly the worker’s conflict cost. Our survey suggests that this form of more sophisticated wage setting is not too common. First, this more sophisticated policy means that default wage offers are fully indexed to inflation, as the firm must continue to keep workers exactly indifferent to conflict when inflation rises. Second, with the sophisticated policy, we would not observe conflict in equilibrium. Neither prediction is supported by our survey.¹³ We therefore summarize firms’ behavior by the growth rate and the inflation indexation of the default wage (α and γ , respectively) and the frequency of “free” catch-up opportunities for workers (λ), which firms may offer to avoid conflict. In Sections 5, we discipline these parameters using responses to tailored survey questions.

Since we focus on the worker’s problem, we set our model in partial equilibrium and the conflict-induced real wage, $w_{i,t}^*$, is exogenous. Moreover, workers are always employed in our setup regardless of their wage (we have assumed that the distribution of wage gaps has non-positive support, meaning firms also have no incentives to fire the worker). These assumptions will be relaxed in the general-equilibrium model in Appendix C.2, which incorporates an employment margin and endogenizes the conflict-induced real wage.

Our model is similar to the standard menu-cost model of price setting (e.g., Golosov and Lucas, 2007; Gertler and Leahy, 2008; Alvarez et al., 2016; Auclert et al., 2024a). However, we apply the model to wage setting and in doing so, impose three important differences. First, in the menu-cost model, the firm’s objective depends on a quadratic loss based on the gap between the current price and the optimal reset price, while in our model, the worker’s objective is always increasing in the gap between the current wage and the conflict-induced wage ($x_{i,t}$). Second, the adjustments in our model are one-sided—workers take actions to raise their wage but not to cut it. In standard menu-cost models, adjustment is two-sided, as firms pay menu costs to either raise or lower prices. Third, in our model, all workers periodically receive wage increases, even without costly conflict and even if they do not receive a free wage catch-up opportunity, due to the default wage increases governed by α and γ . In

¹³Firms might not engage in sophisticated wage-setting policies if there are costs to adjusting wages away from the default wage offer, such as managerial frictions to rearranging contracts.

the standard menu cost model, prices instead remain unchanged if the firm does not pay a menu cost or receive an exogenous free opportunity to change their price.

4.2 The Impact of Inflation Shocks on Worker Welfare and Wages

The economy starts from a steady state with inflation $\pi^{ss} \geq 0$. An unexpected aggregate shock to the path of inflation $\{\hat{\pi}_t \equiv \pi_t - \pi^{ss}\}_{t=0}^{+\infty}$ is realized at the beginning of period 0.¹⁴ The economy does not face other aggregate shocks. We are interested in characterizing how the inflation shock affects workers' welfare and (real) wages. Specifically, workers' aggregate welfare and (real) wages are defined as

$$\mathcal{W} \equiv \int_0^1 \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \{ \log(c_{i,t}) - \kappa_{i,t} \mathcal{J}_{i,t} \} \right] di \quad \text{and} \quad \log w_t \equiv \int_0^1 \log(w_{i,t}) di, \quad (5)$$

where $\mathbb{E}[\cdot]$ averages over the realization of idiosyncratic shocks (after the realization of the aggregate shock). The impact of the inflation shocks on workers' welfare and wages is denoted by $\hat{\mathcal{W}} \equiv \mathcal{W} - \mathcal{W}^{ss}$ and $\{\hat{w}_t\}_{t=0}^{\infty} \equiv \{\log w_t - \log w^{ss}\}_{t=0}^{\infty}$.

To characterize the responses, we first derive the rule characterizing the worker's optimal choice over whether to engage in conflict with their employer. Specifically, we can rewrite the utility of worker i in equation (2) as a function of wage gaps, conflict decisions, and an exogenous constant that is invariant to conflict decisions and the path of inflation:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \{ \log(w_{i,t}^* e^{x_{i,t}}) - \kappa_{i,t} \mathcal{J}_{i,t} \} \right] = \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t (x_{i,t} - \kappa_{i,t} \mathcal{J}_{i,t}) \right] + \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \log(w_{i,t}^*) \right]. \quad (6)$$

Worker i 's problem can then be summarized by:

$$\max_{\{\mathcal{J}_{i,t}\}_{t=0}^{\infty}} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t (x_{i,t} - \kappa_{i,t} \mathcal{J}_{i,t}) \right] \quad \text{s.t.} \quad (4). \quad (7)$$

In each period t , a worker faces two options. First, the worker can choose to conflict with the employer ($\mathcal{J}_{i,t} = 1$), increasing pay and eliminating the wage gap ($x_{i,t} = 0$). Second, the worker can refrain from conflict ($\mathcal{J}_{i,t} = 0$) and allow the wage to adjust according the employer's default wage offer, $x_{i,t} = x_{i,t}^d \equiv x_{i,t-1} - (\mu + z_{i,t}) - (1 - \gamma) \hat{\pi}_t$, where $x_{i,t}^d$ captures the wage gap implied by the employer's default wage

¹⁴Both positive and negative shocks to $\hat{\pi}_t$ are allowed. Because of idiosyncratic shocks, the effects of inflation shocks on aggregate worker welfare are locally symmetric.

offer. When conflict is costly ($\kappa_{i,t} = \kappa$), the worker's optimal conflict choice can be characterized by a threshold rule: there exist thresholds $\{\underline{x}_t\}_{t=0}^{+\infty}$ such that the worker engages in conflict ($\mathcal{J}_{i,t} = 1$) if $x_{i,t}^d \leq \underline{x}_t$ and does not $\mathcal{J}_{i,t} = 0$ if $x_{i,t}^d > \underline{x}_t$. We use $\underline{x}^{ss}$ to denote the steady-state conflict threshold, i.e., the value of x at which the worker is indifferent between conflict with employers and accepting the default wage at steady-state inflation π^{ss} .

In the model, there is conflict at steady state, even without shocks to inflation. At π^{ss} , from the dynamics of wage gap in (4), there are two reasons why a worker's wage gap can be pushed below the conflict threshold $\underline{x}^{ss}$ and induce conflict: first, a positive average drift of the wage gap in steady state $\mu > 0$; and second, a large positive idiosyncratic shock $z_{i,t}$.

We now turn to characterizing the impact of inflation shocks on worker welfare. We first show that inflation increases the fraction of workers engaging in conflict, consistent with the survey and observational evidence in Finding 4 above. We define $\text{frac}_t \equiv \int_0^1 \mathcal{J}_{i,t} di$ as the share of workers who conflict with the employer at each time t .

Proposition 1. *If $\gamma < 1$, then an increase in inflation at $t = 0$ leads to a larger fraction of workers engaging in conflict at $t = 0$, so that $\left. \frac{\partial \text{frac}_0}{\partial \pi_0} \right|_{\{\pi_t = \pi^{ss}\}_{t=0}^{+\infty}} > 0$.*

The intuition for this result is straightforward. Suppose that inflation increases. As long as default wages are not fully indexed to inflation shock ($\gamma < 1$), workers real wages fall absent conflict. As workers' nominal wages fall further behind prices, more workers are pushed over their conflict threshold and choose to incur conflict costs in exchange for higher wages. On the other hand, if default wage offers are fully indexed to inflation shocks ($\gamma = 1$), then real wages do not move with inflation, meaning that workers' conflict decisions do not change as inflation rises. This result occurs because wage setting is state dependent in the model. Alternatively, if wage setting were time dependent—a special case of our model with $\kappa \rightarrow \infty$ and $\lambda > 0$ —conflict $\{\mathcal{J}_{i,t}\}_{t=0}^{+\infty}$ would not change with inflation.¹⁵

We now study the impact of inflation shocks on aggregate worker welfare and how it connects with the responses of aggregate wages. We first decompose the response of aggregate (real) wages to

¹⁵One may wonder how our model differs from the time-dependent model in Erceg et al. (2000). Note that Erceg et al. (2000) focus exclusively on the intensive margin of labor supply adjustment, meaning workers have to accept a lower wage to increase earnings (because the elasticity of labor demand is greater than 1). We instead focus on the extensive margin of labor supply adjustment, meaning workers ask for a higher wage to increase earnings.

inflation shocks into two terms:

$$\hat{w}_t = \hat{w}_t^{\text{erosion}} + \hat{w}_t^{\text{catch-up}}.$$

The first term, which we call *wage erosion*, is the impact of inflation shocks on real wages while holding each worker's conflict decision $\{\mathcal{J}_{i,t}\}_{t=0}^{+\infty}$ as if the inflation were fixed at the steady state level. The second term, which we call *wage catch-up*, captures wage adjustment to inflation shocks arising from the impact of inflation on each worker's conflict decision.

Formally, let $\omega_t(\boldsymbol{\pi}_t, \mathcal{J}_{i,t}, h_{i,t})$ denote worker i 's real wage at time t for a given path of inflation $\boldsymbol{\pi}_t = \{\pi_\tau\}_{\tau=0}^t$, conflict choices $\mathcal{J}_{i,t} = \{\mathcal{J}_{i,\tau}\}_{\tau=0}^t$, and idiosyncratic history $h_{i,t} \equiv (\{z_{i,\tau}, \kappa_{i,\tau}\}_{\tau=0}^t, w_{i,-1}, w_{i,-1}^*)$. Wage erosion measures how aggregate real wages would change in response to inflation shocks, holding conflict decisions fixed at their steady state value:

$$\hat{w}_t^{\text{erosion}} \equiv \int_0^1 \log(\omega_t(\boldsymbol{\pi}_t, \mathcal{J}_{i,t}^{ss}, h_{i,t})) di - \int_0^1 \log(\omega_t(\boldsymbol{\pi}^{ss}, \mathcal{J}_{i,t}^{ss}, h_{i,t})) di. \quad (8)$$

Here, $\mathcal{J}_{i,t}^{ss}$ is what conflict decisions would have been, given steady-state inflation, as well as the same history of idiosyncratic shocks. Wage catch-up is the component of wage adjustment that results from changes in conflict choices due to inflation shocks:

$$\hat{w}_t^{\text{catch-up}} \equiv \int_0^1 \log(\omega_t(\boldsymbol{\pi}_t, \mathcal{J}_{i,t}, h_{i,t})) di - \int_0^1 \log(\omega_t(\boldsymbol{\pi}_t, \mathcal{J}_{i,t}^{ss}, h_{i,t})) di. \quad (9)$$

We can now examine the impact of inflation on aggregate worker welfare. From (5), we can decompose this impact into two components:

$$\hat{\mathcal{W}} = \underbrace{\sum_{t=0}^{\infty} \beta^t \hat{w}_t}_{\text{aggregate wage response}} - \underbrace{\hat{\kappa}}_{\text{aggregate costs of inflation due to conflict}}. \quad (10)$$

The first term captures the effect of inflation on the present value of aggregate wages. The second term, the aggregate costs of inflation due to conflict, is given by

$$\hat{\kappa} \equiv \int_0^1 \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \kappa_{i,t} (\mathcal{J}_{i,t} - \mathcal{J}_{i,t}^{ss}) \right] di = \kappa \sum_{t=0}^{\infty} \beta^t (\text{frac}_t - \text{frac}^{ss}) \quad (11)$$

and captures how the inflation shock changes the total conflict costs borne by workers. This term is equal to the utility cost per conflict action multiplied by how inflation changes the present value of

the fraction of workers who engage in conflict. We now connect the two components of the welfare change to the two components of wage adjustment, and in doing so, present the main analytical result of the paper.

Theorem 1. *The first order impact of inflation shocks $\{\hat{\pi}_t\}_{t=0}^{\infty}$ on aggregate worker welfare is given solely by wage erosion, whereby*

$$\hat{\mathcal{W}} \approx \sum_{t=0}^{\infty} \beta^t \hat{w}_t^{erosion} = \underbrace{\sum_{t=0}^{\infty} \beta^t \hat{w}_t}_{\text{real wage responses}} - \sum_{t=0}^{+\infty} \beta^t \hat{w}_t^{catch-up}, \quad (12)$$

because the welfare gains from wage catch-ups in response to inflation shocks are offset by the associated conflict costs to the first order:

$$\hat{\pi} \approx \sum_{t=0}^{\infty} \beta^t \hat{w}_t^{catch-up}. \quad (13)$$

Equation (12) shows that the response of workers' welfare to inflation shocks depends only on wage erosion. The benefits of wage catch-up equal the conflict costs associated with the wage catch-up, meaning catch-up achieved through costly conflict is irrelevant for worker welfare. The result follows from the envelope theorem of [Milgrom and Segal \(2002\)](#), which applies to discrete choices, i.e., workers' optimal choices of whether to engage in conflict with employers $\{\mathcal{I}_{i,t}\}_{t=0}^{\infty}$. The result also applies to an arbitrary sequence of inflation shocks (including deflationary shocks), regardless of their persistence.

To illustrate the intuition behind this theorem, consider a sequence of positive inflation shocks $\hat{\pi}_t \geq 0$ for all t . First, consider the infra-marginal workers whose conflict decisions are unaffected by inflation shocks (i.e., those workers' wage gaps are above the conflict threshold $\underline{x}^{ss}$). Inflation shocks erode their real wages, and there is no wage catch-up since their conflict decisions do not change with inflation shocks. The impact of inflation shocks on their welfare is hence captured by the wage erosion term, which equals the impact of inflation shocks on their observed real wages.

Second, consider the marginal workers who switch from not engaging in conflict to engaging in conflict due to inflation shocks. These workers are pushed over the conflict threshold as the inflation shocks erode their real wages. Due to worker optimality, before the (small) inflation shock, these marginal workers were near the conflict threshold and were approximately indifferent between engaging in conflict and accepting the employer's offer. As a result, even though these workers experience positive wage catch-up due to conflict, their benefits from wage catch-up are equal to the conflict

costs. Consequently, the impact of inflation shocks on welfare is still captured by the wage erosion term for these marginal workers.

One consequence of Theorem 1 is that the impact of inflation shocks on worker welfare and real wages can differ significantly. Welfare depends solely on wage erosion, while changes in the real wage also reflect wage catch-up achieved through changes in conflict decisions. Even if the aggregate wage keeps up with inflation, meaning that the impact of inflation shocks on the aggregate real wage $\sum_{t=0}^{\infty} \beta^t \hat{w}_t$ is close to zero, inflation could still harm worker welfare. Benefits of wage catch-up resulting from more frequent conflict are offset by the costs associated with conflict. As a result, measuring worker welfare using observed wage growth understates the costs of inflation.

Only in the special case of our model with time-dependent wage setting is the impact of inflation shocks on the aggregate real wage sufficient to capture their impact on worker welfare. In this case, with $\lambda > 0$ and $\kappa \rightarrow \infty$, Theorem 1 still holds. After a sequence of positive inflation shocks, conflict $\mathcal{I}_{i,t} = 1$ if and only if the exogenous “free” catch-up opportunity arrives. As a result, conflict decisions $\{\mathcal{I}_{i,t}\}_{t=0}^{+\infty}$ are invariant to the inflation shocks. Aggregate costs of conflict $\hat{\kappa}$ are always zero, and the wage catch up term $\hat{w}_t^{\text{catch-up}}$ in (9) also equals zero.

The theorem focuses on the first-order impact of inflation shocks on worker welfare. How do large shocks and non-linearity affect our results? Theorem 1 does not hold exactly, because some workers would strictly prefer to engage in conflict, as the shock pushes them over the conflict threshold. However the general lesson remains the same. Workers have to engage in costly conflict to achieve wage gains. Therefore, the impact of inflation shocks on worker welfare differs from the effect of inflation on real wages, and the aggregate cost of inflation due to conflict can be significant. This lesson can be seen from the decomposition of the inflation shock’s impact on worker welfare (10), which does not rely on a first-order approximation. In fact, if the inflation shock becomes very large, the aggregate costs of inflation due to conflict are the primary component of the overall costs of inflation – eventually, everyone engages in conflict, leading to relatively large aggregate costs of inflation due to conflict and relatively small aggregate real wage falls after an inflation shock.

What determines the magnitude of wage erosion, and as such the impact of inflation shocks on worker welfare? The following proposition links wage erosion to two factors: first, the indexation of the default wage; and second, the frequency of conflict at steady-state inflation. Welfare costs are higher if there is less conflict at steady-state inflation or less indexation.

We capture the frequency of conflict at steady-state inflation by the fraction of workers who, at

steady state, do not engage in conflict for k periods, $\Phi_k^{ss} \equiv \int_0^1 \left(\prod_{s=0}^k (1 - \mathcal{J}_{i,t+s}^{ss}) \right) di$ —that is, the probability that the employer’s default wage offer “survives” without conflict for k periods. When conflict is frequent, the survival probability is low. We now state our proposition.

Proposition 2. *To a first order, wage erosion and the impact of inflation shocks on worker welfare is given by*

$$\hat{w}_t^{erosion} \approx -(1-\gamma) \sum_{s=0}^t \Phi_{t-s}^{ss} \hat{\pi}_s \quad \forall t \geq 0 \quad \text{and} \quad \hat{\mathcal{W}} \approx -(1-\gamma) \sum_{s=0}^{\infty} \beta^s \left(\sum_{k=0}^{\infty} \beta^k \Phi_k^{ss} \right) \hat{\pi}_s, \quad (14)$$

The proposition shows that given an inflation shock $\hat{\pi}_s$, wage erosion is a function only of γ , the indexation of the default wage; and the frequency of conflict in steady state as measured by the survival probability Φ_{t-s}^{ss} . Welfare, being the present value of wage erosion, depends on the same two factors. Welfare losses are smaller when indexation is high—in the extreme case of full indexation ($\gamma = 1$), inflation shocks do not lead to any wage erosion. Welfare losses are also smaller when conflict is more frequent at steady-state inflation, meaning survival probabilities Φ_{t-s}^{ss} are low.

Intuitively, imagine a worker who did not engage in conflict between periods s and t . For such a worker, the inflation shock $\hat{\pi}_s$ lowers their real wage at t by $(1-\gamma)\hat{\pi}_s$. The term $1-\gamma$ captures the fact that the inflation shock may not lower real wages one-for-one even without conflict because of indexation—and with full indexation, the inflation shock does not lower real wages at all. Suppose instead that the worker has engaged in conflict between periods s and t . Then, $\hat{\pi}_s$ will not erode their real wage at t , because conflict results in nominal wages that fully catch up with inflation. Summing across workers, $\hat{\pi}_s$ erodes the aggregate real wage at t by $-(1-\gamma)\Phi_{t-s}^{ss}\hat{\pi}_s$. Summing over inflation shocks in all periods, we arrive at the expression for wage erosion in (14). The impact of inflation shocks on worker welfare then follows from equation (12).

The main value of Proposition 2 is clarifying how key parameters of the model impact the welfare costs of inflation. First, a higher conflict cost κ means that workers engage in conflict less frequently at steady state, which raises the survival probabilities of the employer’s default wage offer $\{\Phi_t^{ss}\}$. Therefore, higher conflict costs increase the magnitude of wage erosion and raise the welfare costs of inflation. Similarly, a lower probability of free wage catch-up λ or a higher growth rate of the default nominal wage at steady state α also raises the survival probabilities $\{\Phi_t^{ss}\}$ and, hence, the magnitude of wage erosion and the welfare costs of inflation. Finally, higher indexation γ increases the employer’s default wage offer after an inflation shock, which lowers the magnitude of wage ero-

sion and the welfare costs of inflation. Guided by the proposition, we will provide direct empirical evidence about these parameters in Section 5.

4.3 Extensions

We now consider various extensions of the baseline model. The core result—that the impact of inflation shocks on workers’ welfare is determined by wage erosion and not by real wage growth—continues to apply.

More general distribution of conflict costs. Our main result, Theorem 1, does not depend on the "Calvo-plus" form and holds for a more general distribution of conflict costs with non-negative supports. As further elaborated in Appendix C.1, the application of the envelope theorem in Milgrom and Segal (2002) does not require specific restrictions on the distribution of conflict costs.

Conflict-induced real wages affected by inflation shocks. In our main analysis, if the worker takes actions to increase their pay, their wages exactly keep up with inflation. In other words, the conflict-induced (real) wage $w_{i,t}^*$ is invariant to inflation shocks, i.e., fully indexed. Our main result, Theorem 1, can be extended to the case where $w_{i,t}^*$ is affected by inflation shocks, i.e., partially indexed or overshooting. This could occur because conflict-induced wages depend on labor market tightness (as in Appendix C.2) or due to real rigidities, whereby the conflict-induced wage partly depends on economy-wide wages. In this case, the impact of inflation shocks on aggregate worker welfare is still given by $\hat{\mathcal{W}} \approx \sum_{t=0}^{\infty} \beta^t \hat{w}_t^{\text{erosion}}$, where wage erosion—how aggregate shocks would impact the workers’ real wages if their conflict decisions did not respond to aggregate shocks—is given by:

$$\hat{w}_t^{\text{erosion}} \approx -(1-\gamma) \sum_{s=0}^t \Phi_{t-s}^{ss} \hat{\pi}_s + \sum_{s=0}^t (1-\Phi_{t-s}^{ss}) \hat{g}_{w,s}, \quad (15)$$

where $g_{w,s} \equiv \log(w_s^*/w_{s-1}^*)$ is the growth rate of aggregate conflict-induced (real) wages $\log w_s^* \equiv \int_0^1 \log(w_{i,s}^*) di$ and deviations from their steady-state values are still denoted by hats. The first term is exactly the same as in (14), and the second term captures the impact from changes in the growth of conflict-induced real wages if workers’ conflict decisions are fixed at steady-state value. Intuitively, the growth $\hat{g}_{w,s}$ increases the real wage at t only if workers engage in conflict between periods s and t , with the probability $1 - \Phi_{t-s}^{ss}$, i.e., the probability that the employer’s default wage offer does not “survive” for $t-s$ periods. Similar to the main analysis, wage changes arising from changes in conflict decisions due to inflation shocks still do not impact worker welfare. Appendix C.1 provides details.

Allowing other aggregate shocks. In the main analysis, we study the case in which the only aggregate shocks are inflation shocks. Our main result, Theorem 1, can be extended to the case with other aggregate shocks (e.g., TFP shocks from changing aggregate productivity growth $\hat{g}_t \equiv g_t - g^{ss}$). This case is analogous to the previous case in (15), where the first-order impact of TFP shocks on aggregate worker welfare is still given by wage erosion, which depend both on how TFP shocks impact inflation and how they impact aggregate conflict-induced (real) wages. Appendix C.1 provides details.

This extension clarifies two things. First, the impact of inflation on worker welfare does not depend on the underlying source of inflation. Regardless of whether inflation is caused by TFP shocks $\{\hat{g}_s\}$ or other shocks independent of $\{\hat{g}_s\}$, the extent to which inflation leads to wage erosion and its impact on worker welfare remains the same. Second, when workers need to engage in costly conflict to achieve wage gains, "conflict costs" are relevant for the welfare costs of any other aggregate shocks to which workers' default wages are not fully indexed.

The intensive margin of conflict—allowing conflict costs to scale with wage gains from conflict. In our baseline analysis, conflict costs are fixed and do not depend on the wage gains from conflict (the gap between conflict-induced wage $w_{i,t}^*$ and the default wage $w_{i,t}^d$). Moreover, after engaging in conflict to raise their pay, the worker's wage is given exogenously by $w_{i,t} = w_{i,t}^*$ and is not chosen by the worker. We consider an alternative setup that captures the intensive margin of conflict. Workers choose the conflict-induced wage, and conflict costs increase with wage gains—akin to Rotemberg costs in price setting. Specifically, when engaging in conflict to raise their wages beyond the default offer, the worker chooses their wage $w_{i,t}$ but incurs a period- t utility cost of $\frac{\kappa}{2} \left(\log w_{i,t} - \log w_{i,t}^d \right)^2$. In this case, our main result remains true: the impact of inflation shocks on aggregate worker welfare is still given by $\hat{\mathcal{W}} \approx \sum_{t=0}^{\infty} \beta^t \hat{w}_t^{\text{erosion}}$. Wage erosion is still defined as how inflation shocks would impact workers' real wages if their conflict decisions (defined now in terms of the intensity of the conflict $\log w_{i,t} - \log w_{i,t}^d$) were held at steady-state level. Moreover, the impact of inflation shocks on worker welfare is more simply given by $\hat{\mathcal{W}} \approx -(1-\gamma) \sum_{s=0}^{\infty} \beta^s \sum_{k=0}^{\infty} \beta^k \hat{\pi}_s$, which is as if $\Phi_k^{ss} = 1$ for all k in (14). The result follows because the inflation shock $\hat{\pi}_s$ would lower workers' real wage at $t \geq s$ by $(1-\gamma) \hat{\pi}_s$ if the intensity of their conflict decisions were held at steady-state level (see Appendix C.1). Therefore a given inflation shock lowers worker welfare by more in the extension than in the baseline model.

Beyond hand-to-mouth consumers. In the main analysis, we study the case in which the worker has log utility and is hand-to-mouth. Our main result, Theorem 1, can be extended to the case in which the worker faces a standard borrowing constraint and/or does not have log utility. As elaborated

in Appendix C.1, the impact of inflation $\{\hat{\pi}_t\}_{t=0}^{+\infty}$ on aggregate worker welfare is now given by

$$\hat{\mathcal{W}} \approx \sum_{t=0}^{\infty} \beta^t \left[\int_0^1 u'(c_{i,t}^{ss}) w_{i,t}^{ss} di \right] \int_0^1 \frac{u'(c_{i,t}^{ss}) w_{i,t}^{ss}}{\int_0^1 u'(c_{i,t}^{ss}) w_{i,t}^{ss} di} \hat{w}_{i,t}^{\text{erosion}} di, \quad (16)$$

where $\hat{w}_{i,t}^{\text{erosion}} \equiv \log(\omega_t(\boldsymbol{\pi}_t, \mathcal{J}_{i,t}^{ss}, h_{i,t})) - \log(\omega_t(\boldsymbol{\pi}^{ss}, \mathcal{J}_{i,t}^{ss}, h_{i,t}))$. Compared to (12), there are two differences. First, different workers' wage erosion may receive different weights $\frac{u'(c_{i,t}^{ss}) w_{i,t}^{ss}}{\int_0^1 u'(c_{i,t}^{ss}) w_{i,t}^{ss} di}$: workers with lower levels of consumption have a higher marginal utility of consumption, so they receive a relatively higher weight conditional on their wage. Second, the overall welfare consequences of wage erosion are smaller when workers have more ability to smooth temporary wage fluctuations. However, the envelope theorem logic, where the benefits of wage catch-up to inflation shocks on worker welfare are offset by the costs from more frequent conflict, continues to apply.

5 Using the Survey to Calibrate Key Model Parameters

As we have discussed, several parameters of the model determine the extent to which conflict affects the costs of inflation: how much workers dislike taking actions to get higher pay (i.e., the utility cost to each worker of engaging in conflict κ); the probability of free wage catch-up λ ; the degree of indexation of employers' default wage offers γ ; and the growth rate of the default nominal wage at steady state inflation. This section uses our survey to directly measure the parameters. These survey estimates can then be directly used to discipline key parameters of the model, which will allow us to quantify how conflict affects the costs of inflation in the next section.

5.1 Conflict Costs

We first measure what fraction of their wages workers would sacrifice to avoid conflict with their employers. We do so in two steps. First, we elicit the wage growth workers believe they could secure from their employer if they took costly actions to increase pay. Second, we elicit the default wage growth that makes workers indifferent between accepting the employers' default wage offer versus choosing to take those actions. The difference between these two wages measures the fraction of their wages that workers would give up to avoid conflict with employers.

Figure 4: Survey Question to Elicit Indifference Wage

	I would accept my employer's pay growth offer	I would do my best using any strategies at my disposal to increase my pay further
Employer offers you pay growth of 4%	☐	☐
Employer offers you pay growth of 3.5%	☐	☐
Employer offers you pay growth of 3%	☐	☐
Employer offers you pay growth of 2.5%	☐	☐
Employer offers you pay growth of 2%	☐	☐
Employer offers you pay growth of 1.5%	☐	☐
Employer offers you pay growth of 1%	☐	☐
Employer offers you pay growth of 0.5%	☐	☐
Employer offers you pay growth of 0%	☐	☐

Notes: this figure contains the question from the survey eliciting the indifference wage. For each hypothetical pay growth offered by the employer, respondents are required to choose whether they would accept the offer or take costly actions to increase pay. Respondents first answer a question that reveals their conflict induced wage: “[c]ommon strategies to increase pay include initiating a difficult conversation about pay with employers, or searching for higher paid jobs with other employers. Please, think ahead to 12 months from now. Suppose that you are working at the same job at the same place you currently work, and working the same number of hours. What pay growth do you think you would get if you do your best to increase pay using any strategies at your disposal, including the common strategies listed above?”. In order to elicit indifference wage growth, respondents then answer the question “[y]our employer increases pay for everyone in your position, including you, by z% (possible values listed below). Would you accept your employer’s offer without taking any actions to increase your pay or would you do your best to increase your pay using any strategies at your disposal (such as initiating a difficult conversation about pay with employers, or searching for higher paid jobs with other employers)? Remember that you have said that if you do your best to increase pay using any strategies at your disposal, you would have a pay growth of X%.” Here, X is their answer to the previous question.

To elicit wage growth induced by conflict, we ask respondents to think ahead 12 months, while holding fixed their current work place, employer, job and working hours. We ask the worker what nominal wage growth they think they could achieve if they were to use any strategies at their disposal, $\Delta W^{\text{conflict}}$.

We then elicit the default nominal wage growth at which workers are indifferent between accepting their employer’s offer versus choosing to take costly action, ΔW^{indiff} . The fraction of the wage that workers would sacrifice to avoid conflict is then $x^{\text{conflict}} = \Delta W^{\text{conflict}} - \Delta W^{\text{indiff}}$.¹⁶ As we will discuss shortly, this object directly maps to the conflict threshold $\underline{x}^{\text{ss}} = -x^{\text{conflict}}$, which is tightly linked to the conflict cost κ .

¹⁶Figure 1 reports related information on the wage increase that workers who did not conflict in 2023 would hypothetically receive with conflict. This information bounds, but does not point identify, each worker’s conflict cost.

To elicit ΔW^{indiff} , we adapt the standard “multiple price lists for willingness to pay elicitation” used in experimental economics (e.g., [Jack, McDermott, and Sautmann, 2022](#)). Based on the reported conflict-induced nominal wage growth $\Delta W^{\text{conflict}}$, we constructed a menu of nominal wage growth options where the maximum nominal wage growth is $\Delta W^{\text{conflict}}$ and the minimum is $\Delta W^{\text{conflict}}$ minus 4 percentage points, with a gradient of 0.5 percentage points. Figure 4 shows this menu, for an example in which the respondent reported conflict-induced wage growth $\Delta W^{\text{conflict}} = 4\%$. For each hypothetical wage growth in the menu, we asked participants whether they would accept the offer or take actions to achieve higher pay.¹⁷ In the top row, employers offer wage growth equal to the conflict-induced wage. If conflict is costly, workers should always accept this wage. In the bottom row, employers offer far less wage growth than the conflict induced wage, meaning workers should choose conflict unless the costs are prohibitively high. At some intermediate wage growth hypothetically offered by the employer, workers should switch between accepting and engaging in conflict. The switching wage growth bounds the worker’s indifference wage growth (ΔW^{indiff}) within a 0.5% interval. Specifically, letting $\Delta W^{\text{accept}} \in [\Delta W^{\text{indiff}} - 0.5\%, \Delta W^{\text{indiff}}]$ denote the lowest nominal wage growth where workers accept the employers’ offer, we can find

$$x^{\text{conflict}} \in [\Delta W^{\text{conflict}} - \Delta W^{\text{accept}}, \Delta W^{\text{conflict}} - \Delta W^{\text{accept}} + 0.5\%]. \quad (17)$$

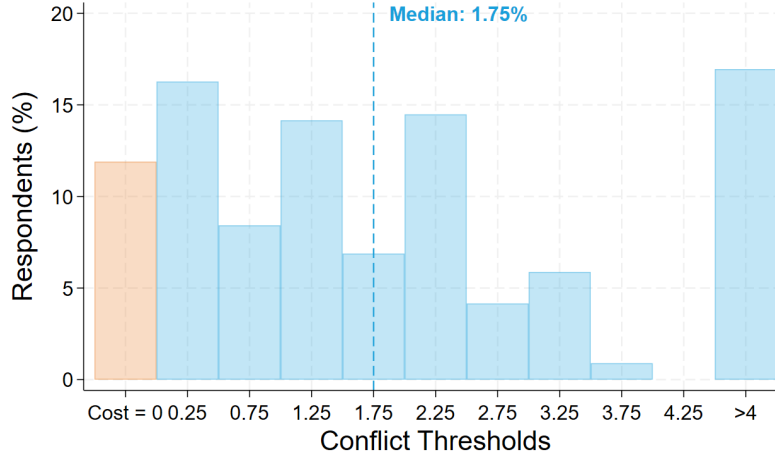
We set x^{conflict} at the median of the interval. For those who would take costly actions at all default wage offers, we assign an x^{conflict} of zero. For those who would never take actions that put them in conflict with their employer, we assume that x^{conflict} is higher than 4%.¹⁸

Figure 5 illustrates the full distribution of conflict thresholds x^{conflict} in our sample. The conflict thresholds are large and heterogeneous. The median worker would sacrifice 1.75% of their wages in order to avoid taking costly actions that put them in conflict with their employer. There is large dispersion around this median value of x^{conflict} , with more than 15% of the sample being willing to sacrifice at least 4 percent of their wages to avoid conflict. While there is substantial dispersion in

¹⁷We randomized whether the menu was ascending or descending, and whether accepting or conflicting is ordered first, which means that the average results are unaffected by any anchoring due to page location. Fortunately, we find that the cost of conflict is the same across groups, meaning order is not important.

¹⁸The only group of respondents that we exclude from Figure 5 are those who give non-monotone responses, which fortunately is less than 7% of the sample.

Figure 5: Distribution of Conflict Thresholds Elicited from Survey



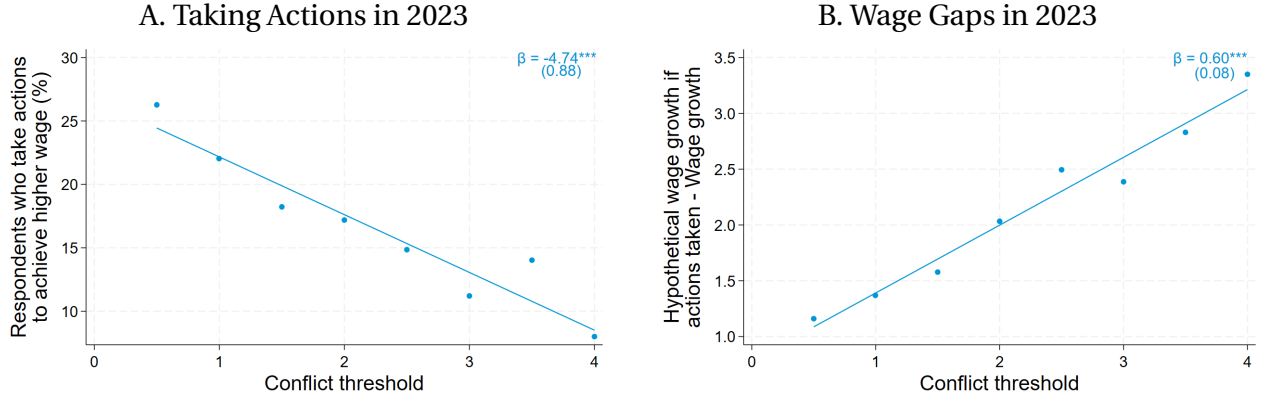
Note: this figure illustrates the distribution of conflict thresholds (x^{conflict} , which maps to $-x^{\text{ss}}$ in the model), showing the percentage of participants with each discrete value. x^{conflict} is defined as the difference between the wage growth participants would receive if they take actions to increase their pay ($\Delta W^{\text{conflict}}$) and their indifference wage (ΔW^{indiff}), the default wage growth at which workers are indifferent between accepting their employer's offer versus choosing to take costly action. Specifically, we set x^{conflict} at the median of the interval in (17). The data excludes respondents who give non-monotonic responses. The median conflict threshold, considering all individuals except participants who always bargain and thus have a cost of zero, is highlighted in the figure.

what workers are willing to sacrifice to avoid conflict, we do not find much systematic heterogeneity across worker demographics or income (See Appendix Figure B.2.) One limitation of this exercise is that other margins of adjustment, beyond wages, may also affect worker utility—for instance vacation days, work from home options, or other amenities.¹⁹

Our measure of x^{conflict} is derived from hypotheticals, but reassuringly, the cross sectional variation is consistent with respondents' self reported actions in 2023. First, we show that respondents who are willing to sacrifice more to avoid conflict were indeed less likely to engage in conflict. We see this in the left panel of Figure 6, where there is a strong negative relationship between the probability

¹⁹To assess magnitudes, one useful comparison is union dues, which approximate how much workers pay to avoid direct conflict with employers. Union dues are generally between 1-2% of wages per year. For example, dues for the Service Employees International Union (health care, 1.9 million members) were 1.7%, and United Auto Workers (auto manufacturing, 1 million members) were approximately 1.1%. Relatedly, as shown in Appendix Table B.2, the only group with significantly lower conflict costs is union members—excluding this group would further increase the aggregate costs of inflation due to conflict.

Figure 6: Validating Elicited Conflict Thresholds



Note: Panel A shows the relationship between the conflict threshold (x^{conflict}) and an indicator for whether the respondent took actions to achieve wage growth in 2023. x^{conflict} is defined as the difference between the wage growth participants would receive if they take actions to increase their pay ($\Delta W^{\text{conflict}}$) and their indifference wage (ΔW^{indiff}), which is defined as the minimum wage growth participants would be willing to accept if offered by their employers. The data is limited to respondents who bargain first and then accept the offer. The coefficient of this relationship is displayed, with the standard errors enclosed in brackets. Stars denote levels of statistical significance: 1% (***), 5% (**), and 10% (*). The binned scatterplot in Panel B shows the relationship between the cost of conflict (x^{conflict}) and the difference between the wage respondents think they could have gotten if they had taken action and the wage growth they received in 2023. The sample is restricted to respondents who accepted their employers' default wage during 2023.

that the respondent took actions in 2023 on the y-axis and their elicited x^{conflict} on the x-axis. Second, we further find a tight link between the wages workers are willing to sacrifice to avoid conflict and the size of workers' perceived wage gains from conflict in 2023. Specifically, the right panel of Figure 6 restricts to those workers who did not engage in conflict in 2023 and relates the wages they were willing to give up to avoid conflict (x^{conflict}) to the wage gain that those workers think they could have achieved if they had taken action (i.e. the data displayed in the right panel of Figure 1). The positive relationship is exactly what we expect – the workers who did not take action in 2023 even when those actions would have resulted in large wage gains are precisely those who had large costs. Conversely, the workers who did not choose to take actions despite low conflict thresholds are precisely those who did not stand to gain as much from those actions. Together, these cross-sectional patterns show consistent answers across sections of the survey and increase our confidence that the elicited conflict costs predict worker behavior.

5.2 Default Nominal Wage Growth and Inflation

The framework in Section 4 shows that the aggregate costs of inflation due to conflict depend on how much the default wage adjusts with inflation, captured by γ . This object is hard to measure

in observational data, since one cannot easily distinguish between conflict-induced wage growth and default wage growth offered by an employer. Instead, we elicit the degree of indexation of default wage offers to inflation using survey hypotheticals. Similar to Section 3, we randomly assign participants into a hypothetical scenario in which inflation is expected to be 2%, 4%, 6%, 8% or 10% over the next 12 months. We then asked survey respondents what nominal wage growth employers would offer them in that setting (i.e. the “default” nominal wage growth).

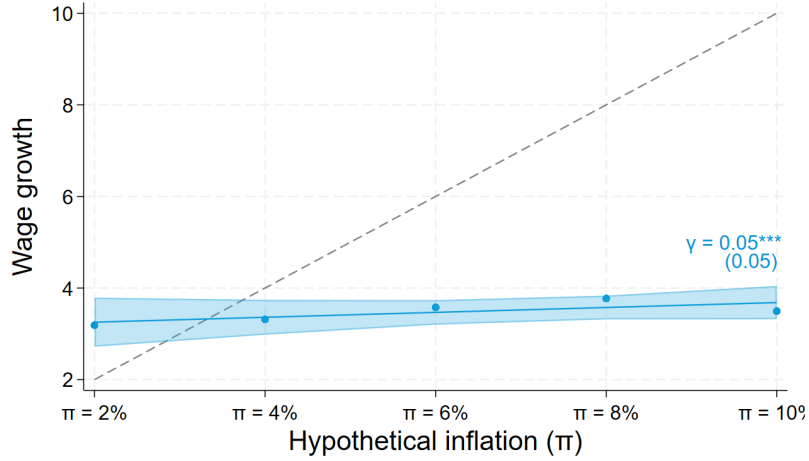
Figure 7 shows that workers perceive that employers would offer almost the same default nominal wage growth at all levels of inflation. The y-axis is default wage growth that workers expect their employer to offer in the absence of any actions on their part. The x-axis is hypothetical inflation proposed to the respondent. For reference, we also plot the 45 degree line, which would reflect fully indexed wages. The blue circles plot the mean default wage growth expected by the respondents in the scenario that was posed to them. The regression line, with shaded 95% confidence intervals, has a slope of 0.05, but is not statistically different from zero. Therefore, workers believe that a 1 percentage point increase in inflation leads to a 0.05 percentage point increase in employers’ default wage offers, absent conflict. Reassuringly, Hajdini et al. (2023) found a similar result—that workers believe higher inflation does not pass through into wages—based on information treatments about the inflation outlook.²⁰ Finally, Figure 7 also shows that workers perceive employers would offer them a default nominal wage annual growth of 3.25% at steady-state inflation (2% annually), which directly maps to α in the model.

5.3 Mapping Survey Evidence to Model

We now use these survey estimates to calibrate the key parameters of the model governing the aggregate costs of inflation due to conflict. To identify conflict costs κ , we follow two steps. First, as discussed in Section 4, with a wage gap of $\underline{x}^{ss} < 0$ at the conflict threshold, the worker is indifferent between engaging in conflict and accepting the default wage offer. In other words, $-\underline{x}^{ss}$ captures the percent of wages that workers would sacrifice to avoid conflict in the model, exactly mapping to $-\underline{x}^{ss} = x^{\text{conflict}}$ in the survey. Therefore we set $\underline{x}^{ss} = -1.75\%$ in the model. In the second step, we then

²⁰We find similar results when relating the default wage offer that workers think their employer will offer them next year to what they expect inflation will be next year (Appendix Figure B.8).

Figure 7: Default Nominal Wage Growth and Inflation



Note: This binned scatterplot depicts the relationship between the default wage growth respondents reported they would receive under a hypothetical inflation scenario and the hypothetical inflation rate, along with the 95% confidence interval of the predicted relationship. The gray dashed line serves as a reference 45-degree line. The coefficient of this relationship is displayed, with the standard errors enclosed in brackets. The stars indicate levels of statistical significance: 1% (***), 5% (**), and 10% (*). The sample is all respondents. Respondents answered the following question. “Consider a hypothetical situation in which inflation is expected to be $x\%$ in the next 12 months. Suppose that you are working at the same job at the same place you currently work, and working the same number of hours. What pay growth do you think you would get by default if you do not take any strategies at your disposal to increase your pay (such as initiating a difficult conversation about pay with employers, or searching for higher paid jobs with other employers)?” We hypothetically offer inflation of $x = 2\%, 4\%, 6\%, 8\%$ and 10% .

further use $\underline{x}^{ss}$ to pin down the conflict cost parameter κ .²¹

Furthermore, 11% of workers report that they would always engage in conflict over the next year, irrespective of the employer’s offer (the leftmost orange bar in Figure 5). These workers perceive that conflict is costless, which maps to the probability of a free wage catch-up in our model λ . In our quarterly calibration, we translate this yearly share into a quarterly free wage catch-up opportunity with probability $\lambda = 1 - (1 - 0.11)^{1/4} = 2.87\%$.²²

We calibrate γ , the extent to which default wage growth indexes to inflation, to $\gamma = 0.05$ based on the slope of the regression line in Figure 7, i.e., a small degree of indexation of the default wage. This

²¹Specifically, let $v^{ss}(x)$ denote the worker’s value given its end-of-period wage gap in (7) at steady state. That is, $v^{ss}(x) \equiv x + \max_{\{\mathcal{J}_{i,t}\}_{k=1}^{\infty}} \mathbb{E}_t [\sum_{k \geq 0} \beta^{t+k} (x_{i,t+k} - \kappa_{i,t+k} \mathcal{J}_{i,t+k})]$, subject to (4) and $\pi_t = \pi^{ss}$ for all t . One can then use $\underline{x}^{ss}$ to pin down the conflict cost parameter κ : $v^{ss}(0) - \kappa = v^{ss}(\underline{x}^{ss})$.

²²As discussed above, the free wage catch-up opportunity can also arise from firms unsolicitedly offering workers wage increases to avoid conflict. In our numerical robustness checks, we directly explore calibrations of λ based on this interpretation.

regression captures the extent of perceived default wage indexation, which could be lower than actual default wage indexation. In our numerical exercises, we will also explore robustness to much higher levels of γ than our baseline survey estimates.

A last important parameter is α , the growth rate of the default nominal wage at steady state inflation. We also calibrate α from our survey based on hypothetical inflationary scenarios. Specifically, Figure 7 reveals the annual growth rate of default nominal wages at steady state inflation is 3.25%, which implies a quarterly value of $\alpha = 0.81\%$. This calibration implies workers receive periodic default wage increases at steady-state inflation absent conflict.

Two points merit further clarification. First, the elicitation of these key parameters is based on survey hypotheticals that are independent of the specific time the survey is conducted and unrelated to the respondent's actual actions in 2023 that we used to motivation in Section 3. Second, our survey measure elicits conflict costs directly. One could instead infer these conflict costs indirectly, by calibrating a model to match moments of the wage growth distribution as in the menu cost literature (e.g., Alvarez et al., 2016). In our model, wage changes can arise both with and without conflict—for instance, from growth in the default wage offered by employers. Without hard-to-verify assumptions, one cannot differentiate conflict-induced wage changes from default wage changes in the wage data, nor map the distribution of wage changes to conflict costs.

6 Quantifying How Conflict Affects the Costs of Inflation

In this section, we analyze the quantitative implications of the model developed in Section 4 and calibrated in Section 5. The aim is to quantitatively assess the importance of conflict for the costs of inflation to workers. We find that the costs of inflation, including conflict, are significantly larger than the costs of inflation via falling real wages alone.

6.1 Calibration

We calibrate the model from Section 4 at a quarterly frequency and summarize the parameters in Table 1. As we discussed above, we use our survey to measure the costs of conflicts ($\lambda = 2.87\%$ and $\kappa = 8.14\%$ such that $\underline{x}^{\text{ss}} = -1.75\%$), the degree of indexation of the default wage to inflation shocks ($\gamma = 0.05$), and the growth rate of the default nominal wage at steady state inflation ($\alpha = 0.81\%$). Furthermore, for the quarterly calibration, we set the discount factor to a standard value, $\beta = 0.99$. The

trend productivity growth rate $g = 0.76\%$ is chosen to map a steady-state average annual growth rate of real wages of 3.02% in the ASEC-CPS survey.²³ We assume $\pi^{ss} = 0.5\%$, implying a steady-state annual inflation of 2%, again a standard value.

We assume that the idiosyncratic productivity shock is such that $z_{i,t} + \mu$ follows a Gamma(a, b) distribution, a flexible continuous distribution with support $[0, \infty)$. Setting the lower bound of the idiosyncratic productivity shock to $\underline{z} = -\mu$ ensures that the steady-state distribution of wage gaps has a non-positive support. We calibrate the shape and scale parameters of the Gamma distribution $(a, b) = (0.14, 0.03)$ such that idiosyncratic shocks have a mean of zero, $\mathbb{E}[z_{i,t}] = 0$, and the yearly share of workers engaging in conflict at steady-state inflation is equal to 48%, as in Figure 3.²⁴ As in standard menu cost models, the distribution of idiosyncratic shocks impacts how workers' wage gaps move and thus the frequency of conflict. As a result, the share of workers engaging in conflict at steady state inflation is informative about the distribution of idiosyncratic productivity shocks, which in turn impacts the costs of inflation shocks through Proposition 2.

6.2 Quantifying the Aggregate Costs of Inflation Due to Conflict

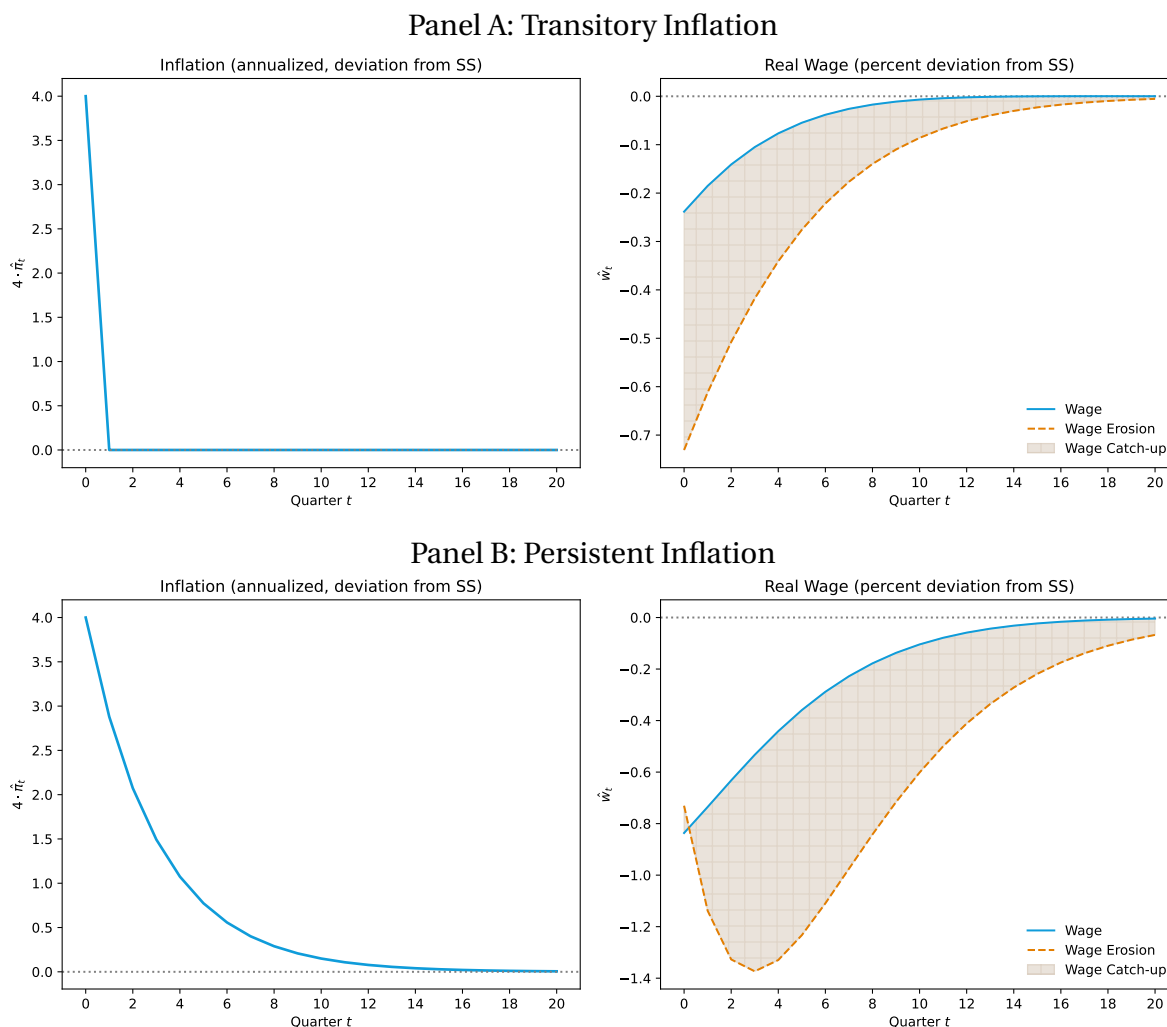
We use the calibrated model to quantify the welfare costs of inflationary shocks to workers. We solve the first-order impact of inflation shocks $\{\hat{\pi}_t\}_{t=0}^{+\infty}$ using Sequence-Space Jacobian methods, developed in Auclert et al. (2021) and Auclert et al. (2024a). This approach allows us to analyze the welfare consequences of an arbitrary path of inflation.

We consider two simple inflationary scenarios in this subsection. The first scenario is a transitory inflation shock (Figure 8, Panel A), in which we set $\hat{\pi}_0 = 1\%$ and $\hat{\pi}_t = 0$ for $t \geq 1$. The second scenario is

²³We access ASEC-CPS data from the IPUMS CPS database (Flood, King, Rodgers, Ruggles, Warren, Backman, Chen, Cooper, Richards, Schouweiler, and Westberry, 2023).

²⁴This choice is conservative in terms of aggregate costs of inflation due to conflict. Alternatively, we could calibrate to the 21% of workers who chose conflict in 2023 (Appendix Figure B.1). Our current choice is conservative because it implies that a high share of workers already engage in conflict each year in steady state, resulting in smaller aggregate costs of the inflation shock due to conflict. The difference in the fraction of workers between Figure 3 and Appendix Figure B.1 can arise because conflict in practice may be less frequent than in hypothetical questions, as workers might underestimate the large monetary, time, and psychological costs of conflict. Additionally, workers who already engaged in conflict in 2022, when inflation peaked, were less likely to engage in conflict again in 2023.

Figure 8: Real Wage Dynamics and the Aggregate Costs of Inflation due to Conflict



Notes: each panel plots the response to a given inflation scenario. In Panel A, there is a transitory shock to inflation lasting one quarter. In Panel B, there is a persistent shock, that decays at quarterly rate $\rho = 0.72$. In the left figure of each panel, we plot the path of annualized inflation after subtracting the steady state inflation based on the historical mean. In the right panel, we plot the percent deviation of the real wage from the steady state in the solid blue line. We also plot wage erosion in the dashed orange line, which captures the impact of inflationary shocks on worker welfare. The gap between the two lines, shaded in gray, represents wage catch-up achieved through more frequent conflict.

Table 1: Main Analysis—Calibration

	<i>Description</i>	<i>Value</i>	<i>Target</i>
β	Discount factor	0.99	Standard
κ	Conflict cost	8.14%	Own survey
λ	Probability of free catch-up	2.87%	such that $\underline{x}^{ss} = -1.75\%$
g	Trend real wage growth	0.76%	Own survey
α	Default nom. wage growth at steady state inflation	0.81%	ASEC-CPS
γ	Indexation of default nominal wage	0.05	3.02% annual real wage growth
π^{ss}	Steady state inflation	0.5%	Own survey
$z_{i,t}$	Idios. shocks $z_{i,t} + \mu \sim \text{Gamma}(a, b)$	(0.14, 0.03)	2% annual inflation
			$\mathbb{E}[z_{i,t}] = 0$
			48% yearly share of conflict

a persistent inflation shock (Figure 8, Panel B), in which we set $\hat{\pi}_t = \rho^t$ with $\rho = 0.72$, which matches the empirical auto-correlation of inflation in the US time series (specifically, the auto-correlation of quarterly PCE inflation between 2013Q1 and 2024Q1). As in Section 4, an unexpected aggregate inflation shock is realized at $t = 0$, and workers have perfect foresight about the path of inflation afterwards.

The right figure of each corresponding panel shows the dynamic response of real wages and wage erosion under each scenario, with the latter being relevant for welfare. For each scenario, the solid blue line displays the overall real wage response $\{\hat{w}_t\}_{t=0}^{+\infty}$.²⁵ The dashed orange line displays the resulting wage erosion $\{\hat{w}_t^{\text{erosion}}\}_{t=0}^{+\infty}$. The shaded region captures the gap between the two, i.e., wage catch-up due to more frequent conflict $\{\hat{w}_t^{\text{catch-up}}\}_{t=0}^{+\infty}$. The present value of wage catch-up measures the aggregate costs of inflation due to conflict $\hat{z}$, defined in (11).

For the transitory inflation shock, real wages fall by around 0.25% on impact. However, the welfare-relevant wage erosion falls by more than 0.7%, meaning that the modest fall in wages is mostly because workers engage in conflict more frequently to raise their wages. For the persistent inflation shock with $\rho = 0.72$, real wages and welfare-relevant wage erosion fall roughly equally on impact. However wage erosion decreases significantly more than real wages in subsequent quarters. Therefore, there is a large but delayed wage catch-up to the inflation shock through more frequent conflicts in later quarters. The delay occurs because workers have incentives to defer conflict when the infla-

²⁵The overall real wage response $\{\hat{w}_t\}_{t=0}^{+\infty}$ can be positive in response to positive inflationary shocks. The reason is that positive inflation shocks lead to more conflict and nominal wage increases, which helps workers' wages catch up not only with inflation shocks but also with steady state inflation.

Table 2: Decomposing the Impact of Inflation Shocks on Worker Welfare

	Overall Welfare Change	Real Wage Response	Aggregate Costs of Inflation due to Conflict
Transitory inflation	−0.95%	−0.22%	−0.73%
Persistent inflation	−3.31%	−1.16%	−2.15%

Notes: the first column shows the overall decline in welfare after the transitory inflation shock (row 1) and the persistent inflation shock (row 2), as a percent of annual consumption. The second column shows the response of present value of real wages in each scenario, again as a percent of annual consumption. The final column shows the response of the aggregate costs of inflation due to conflict $\hat{\pi}$, again as a percent of annual consumption.

tion shock is persistent. Even if they engage in conflict now and their wages keep up with the initial inflation, persistent inflation shocks still cause their wages to fall behind prices in the absence of future conflict. As a result, workers can economize on conflict costs by delaying conflict until inflation has accumulated. Overall, in both cases, the shaded region between real wages and wage erosion is large, meaning a substantial fraction of the wage growth was achieved through costly conflict.

Table 2 confirms the importance of conflict for the welfare costs of inflation. The table displays the welfare costs of inflation to workers and decomposes them into real wage responses and the aggregate costs of inflation due to conflict, according to equation (12). In the table, welfare units are denoted in terms of percent of annual consumption. For all inflationary scenarios, we find that the costs of inflation to workers, considering conflict costs, are more than twice the costs that arise from falling real wages alone. For example, for the persistent inflation shock with $\rho = 0.72$, the overall welfare costs of inflation for workers are 3.31%. The aggregate costs of inflation due to conflict are 2.15%, constituting 65% of the total costs.²⁶ For the transitory shock, aggregate costs of inflation due to conflict constitute 75% of the total costs, an even higher proportion.

²⁶These results are not comparable to quantitative exercises from menu-cost models of price setting, which typically study the welfare costs of steady-state changes in inflation and use permanent decreases in consumption as welfare units (e.g., Nakamura et al., 2018). We instead study the welfare costs of transitory inflation shocks and use decreases in a single year of consumption as welfare units.

6.3 Quantitative Robustness

We first explore the robustness of our findings to the calibration of several key parameters in our model. Here, we discuss results for the persistent inflation shock with $\rho = 0.72$ (displayed in Appendix Figure B.9). Appendix Figure B.10 also presents the analogous results for the transitory shock. Our main lesson—that conflict significantly raises the overall costs of inflation—is robust across an array of alternative calibrations.

Indexation parameter γ . The degree of inflation indexation of default wages is an important input to our calculations. If default wages were fully indexed, then inflation would never affect real wages or conflict decisions. In the baseline analysis, we calibrate a default wage almost without indexation ($\gamma = 0.05$), based on the estimates of Section 5.2. If we instead allow the default wage to be more highly indexed, then the overall costs of inflation to workers fall.

However, while the degree of indexation affects the overall costs of inflation, it does not affect the *relative* importance of conflict. Intuitively, the impact of inflation shocks on worker welfare is proportional to $1 - \gamma$, i.e., all that matters for workers' decisions is the component of inflation that is not automatically accounted for through wage indexation, $(1 - \gamma) \hat{\pi}_t$. The degree of indexation simply scales up both the overall costs of inflation (see, e.g., equation (14)) and aggregate costs of inflation due to conflict, but does not affect the relative importance of conflict. We formalize this intuition:

Proposition 3. *The ratio of aggregate costs of inflation due to conflict, to the overall costs of inflation, $\hat{\kappa}/\hat{W}$, is invariant to the degree of indexation of default wage $\gamma \in [0, 1)$.*

Panel A of Appendix Figure B.9 quantitatively investigates how the indexation parameter affects the impact of the persistent inflation shock. In the left figure of Panel A, we plot the decline in the present value of real wages (blue solid line), and the present value of welfare-relevant wage erosion (dashed orange line), as the degree of indexation γ varies between 0 and 0.6. The upper bound is conservative: the indexation of actual wages to inflation estimated by Smets and Wouters (2007) is 0.58. The indexation of default wages to inflation shocks in the absence of conflict must be lower. The shaded area between the two lines measures the aggregate conflict costs of inflation. Even with the high indexation of 0.6, we still find meaningful conflict costs of inflation. Consistent with Proposition 3, the right figure verifies that the ratio of aggregate conflict costs of inflation to its overall costs remains constant, around 0.6, even as the degree of indexation varies.

Probability of free wage catch-up λ . A second important parameter is λ , the probability that a

worker receives a free wage catch-up without paying conflict costs. We calibrated $\lambda = 2.87\%$ using the share of workers who reported zero conflict costs in our survey, implying a 11% share of annual free catch-up. This strategy could understate the true value of λ . Workers who do not have zero conflict costs might still receive free wage catch-up, if the firm offers workers unsolicited wage increases to avoid conflict. Panel B in Appendix Figure B.9 shows that our conclusions are similar with higher values of λ than the baseline calibration. As we increase λ and hence the share of annual free catch-up, $1 - (1 - \lambda)^4$, wage catch-up from more frequent costly conflict indeed plays a smaller role in determining the decline in the present value of real wages. However, the ratio of aggregate costs of inflation due to conflict to its overall costs remains above 40% even if $\lambda = 12\%$, implying a 40% share of annual free catch-up (i.e., 4 times more than our baseline calibration). External estimates of how wages increase beyond the default wage are well within this range—for instance, Faberman, Mueller, Şahin, and Topa (2022) estimate that only around 4% of workers receive unsolicited job offers each year.²⁷

Conflict costs κ . Finally, we assess the quantitative robustness of our results to variations in the conflict cost κ , which map one-to-one with the conflict threshold $\underline{x}^{ss}$ as explain in Section 5. In our baseline calibration, we calibrate $\kappa = 8.14\%$ so that the conflict threshold is $\underline{x}^{ss} = -1.75\%$, following our survey evidence. Panel C in Appendix Figure B.9 varies the conflicts cost κ such that the conflict threshold $\underline{x}^{ss}$ varies between -2% and -1% , the interquartile range of our estimates. We find that our quantitative results are robust to different parameterizations of conflict costs. Even with $\underline{x}^{ss} = -1\%$, implying a significantly lower conflict cost $\kappa = 3.82\%$, the ratio of aggregate costs of inflation due to conflict to overall costs of inflation remains above 50%.

In Appendix Figures C.2 and C.3, we also illustrate the quantitative robustness of our results to variations in the growth rate of the default nominal wage at steady state inflation α . Appendix Figures C.4-C.6 show how robust the time profile of real wage and wage erosion responses to the persistent inflation shock are to alternative calibration of key parameters.

General equilibrium determination of employment and wages. Our baseline model quantifies the aggregate costs of inflation due to conflict in a setting where all workers are employed and the conflict-induced real wage is exogenous. However, inflation shocks can also increase overall employment in general equilibrium by “greasing the wheels of the labor market” (Blanco and Drenik, 2023). This channel benefits aggregate worker-welfare through both higher employment rates themselves and their upward pressure on wages in general equilibrium. In Appendix C.2, we extend our model

²⁷For each λ , we re-calibrate κ so that $\underline{x}^{ss} = -1.75\%$ and fix all other parameters as in Table 1.

to consider the importance of conflict costs when employment and wages are determined in general equilibrium. In Appendix Figure C.1, we find that, even in this extended setting, aggregate costs of inflation due to conflict remain significant, both in absolute value and as a share of the overall costs of inflation.

6.4 Conflict and the Costs of the Post Pandemic Inflation of 2021-2023

We now return to the baseline model and quantify the importance of conflict in the post-pandemic inflation of 2021–2023. Specifically, we study inflation shocks (the left figure in Panel A of Figure 9) based on the difference between the headline Personal Consumption Expenditures (PCE) Inflation between January 2021 and December 2023 (quarterly frequency, but annualized), and steady-state inflation 2%. Following December 2023, we assume that inflation converges back to steady state at a rate of $\rho = 0.72$. A key difference between this episode and the previous simpler inflation scenarios, is that inflation peaks several quarters after the initial quarter. Therefore, inflation expectations can meaningfully affect the timing of conflict decisions.

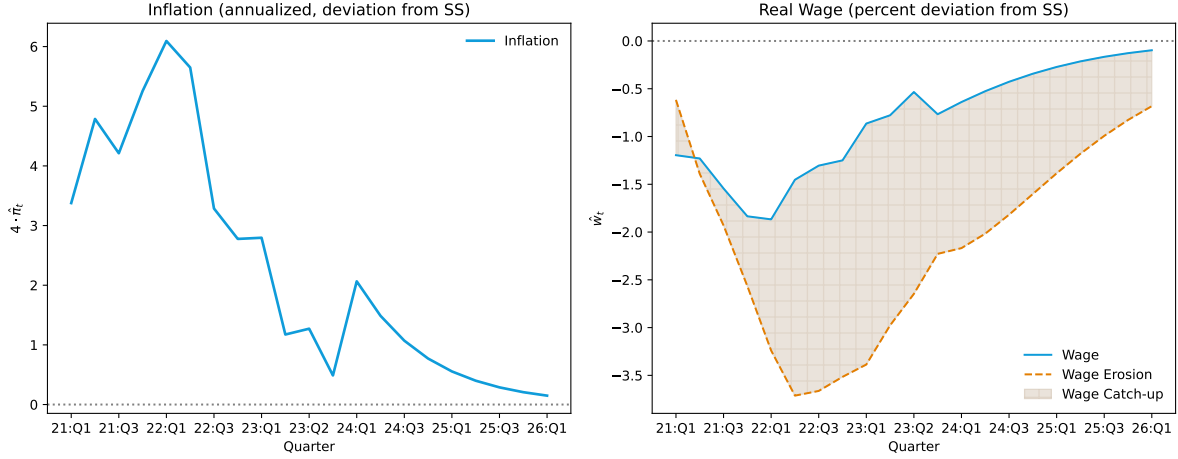
In Panel A of Figure 9, we consider the case of perfect foresight. That is, as in Section 4 and the previous subsection, an unexpected aggregate inflation shock is realized at $t = 0$ (the first quarter of 2021), and workers have perfect foresight about the path of inflation afterwards. In Panel B of the Figure 9, we consider the case where workers' inflation expectations are based on observed inflation expectations in the survey data, by combining the Survey of Professional Forecasters (SPF) and the New York Federal Reserve's Survey of Consumer Expectations (SCE).^{28,29} These inflation expectations are summarized in the right figure in Panel B of Figure 9. In Appendix Figure C.11, we instead assume that workers have no foresight; they expect inflation to be at the steady state in the next period and

²⁸As shown in Auclert, Rognlie, and Straub (2020), the Sequence-Space Jacobian approach can be easily extended to alternative models of expectations. Here, we follow Bardóczy and Guerreiro (2023) and specify workers' inflation expectations directly based on survey data.

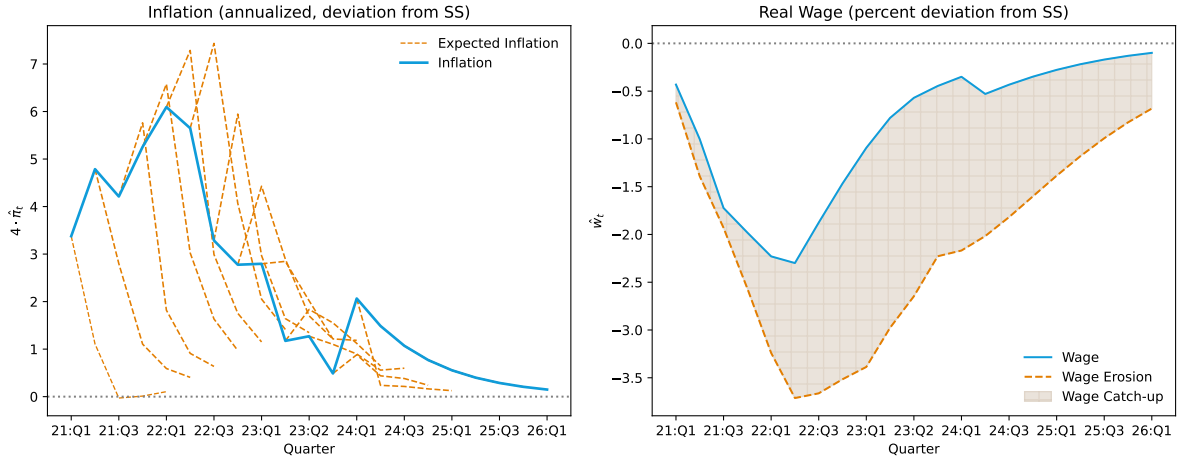
²⁹The SCE provides households' expectations for inflation over the coming 12 months, but does not provide quarterly or long term expectations over this period. The SPF does provide quarterly and long term expectations, but arguably professionals' expectations are less relevant than households'. We create an expectation series that rescales the value of the SPF so that mean expectations over the first year are the same as households' expectations from the SCE. We also use a spline to interpolate medium term expectations, which are not reported in the SPF.

Figure 9: The Aggregate Costs of Inflation due to Conflict during the 2021-2023 Inflation

Panel A: Perfect Foresight



Panel B: Observed Inflation Expectations Based on Survey Data



Notes: In the left figure of Panel A, we plot the path of annualized headline PCE inflation over 2021–2023, after subtracting the steady state inflation based on the historical mean inflation. In the left figure of Panel B, we contrast this realized path of inflation with expected inflation for each quarter, where expected inflation is from the Survey of Consumer Expectations and the Survey of Professional Forecasters, as discussed in footnote 29. In the right figure of both panels, we plot the percent deviation of the real wage from the steady state in the solid blue line. We also plot wage erosion in the dashed orange line, which captures the impact of inflationary shocks on worker welfare. The gap between the two lines, shaded in gray, represents wage catch-up achieved through more frequent conflict. In Panel A, workers have perfect foresight about the path of inflation shocks. In Panel B, workers' inflation expectations are based on observed expectations.

are surprised by inflation in each period.³⁰

The right figure of each corresponding panel in Figure 9 shows the dynamic response of real wages and welfare-relevant wage erosion. Overall, the gap between real wages and wage erosion, which measures the aggregate costs of inflation due to conflict, is large. This is further illustrated in the first and second rows in Appendix Table B.3, which display the welfare costs of inflation to workers and decompose them into real wage responses and the aggregate costs of inflation due to conflict. No matter the assumptions on inflation expectations, the share of conflict remains substantially higher than 50% of the overall welfare losses from inflation (61% under perfect foresight and 59% under observed inflation expectations) for the 2021-23 inflation surge.

Zooming in further, different inflation expectations affect the timing of workers' conflict decisions. As the previous section explains, workers postpone conflict when they expect higher future inflation. However, the envelope theorem implies that changes in conflict decisions due to varying inflation expectations do not impact welfare, as the extra conflict costs paid by workers to secure higher wages exactly cancel out the benefits of those higher wages. Consequently, the overall welfare costs of inflation for workers remain the same in both cases, independent of workers' inflation expectations (The two cases share the same path of welfare-relevant wage erosion (dashed orange lines) in the right figure of each panel, despite having different paths of real wage responses (solid blue lines).)³¹

7 Conclusion

Why do workers dislike inflation so much? We show that “conflict costs” play a significant role: workers must incur these costs to have their nominal wages keep up with inflation, as employers do not automatically provide wage increases when inflation is high. We capture the conflict cost in a menu-cost style model applied to wage setting, and show both analytically and quantitatively, that conflict costs change how inflation shocks impact workers' welfare. Disciplined by a survey of U.S. workers,

³⁰In the case without perfect foresight, to maintain comparability with the perfect-foresight case, we base our welfare assessments on the ex-post realized outcomes; i.e., we use the realized path of inflation to evaluate welfare. An alternative would be to consider “ex-ante welfare” based on workers' expectations.

³¹In Appendix Figure C.7-C.10, we illustrate the robustness of our results for the post-pandemic inflation of 2021–2023 to alternative calibration of key parameters.

we find that incorporating conflict costs substantially increases the costs of inflation to workers.

Beyond the specific application to the costs of inflation, our conflict cost model offers a tractable approach to introducing state-dependent wage setting, providing many avenues for future research. For example, firms may also face costs in adjusting wages away from the default wage offer. These firm-side conflict costs are particularly relevant for downward wage rigidity, as firms would prefer to adjust wages downward when possible. In subsequent work, we aim to quantify these firm-side conflict costs, link them to empirical evidence on downward wage rigidity, and study their macroeconomic implications.

References

- AFROUZI, H., A. BLANCO, A. DRENİK, AND E. HURST (2024a): “A Theory of How Workers Keep Up with Inflation,” *NBER Working Paper*.
- AFROUZI, H., A. DIETRICH, K. MYRSETH, R. PRIFTIS, AND R. SCHOENLE (2024b): “Inflation preferences,” *NBER Working Paper*.
- ALTIG, D., A. AUERBACH, E. EIDSCHUN, L. KOTLIKOFF, AND V. YE (Forthcoming): “Inflation’s Fiscal Impact on American Households,” *NBER Macroeconomics Annual*, 40.
- ALVAREZ, F., M. BERAJA, M. GONZALEZ-ROZADA, AND P. A. NEUMEYER (2019): “From hyperinflation to stable prices: Argentina’s evidence on menu cost models,” *The Quarterly Journal of Economics*, 134, 451–505.
- ALVAREZ, F., H. LE BIHAN, AND F. LIPPI (2016): “The Real Effects of Monetary Shocks in Sticky Price Models: A Sufficient Statistic Approach,” *American Economic Review*, 106, 2817–2851.
- ALVAREZ, F., F. LIPPI, AND A. OSKOLKOV (2022): “The macroeconomics of sticky prices with generalized hazard functions,” *The Quarterly Journal of Economics*, 137, 989–1038.
- AUCLERT, A. (2019): “Monetary Policy and the Redistribution Channel,” *American Economic Review*, 109, 2333–2367.
- AUCLERT, A., B. BARDÓCZY, AND M. ROGNLIE (2023): “MPCs, MPes, and multipliers: A trilemma for New Keynesian models,” *Review of Economics and Statistics*, 105, 700–712.
- AUCLERT, A., B. BARDÓCZY, M. ROGNLIE, AND L. STRAUB (2021): “Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models,” *Econometrica*, 89, 2375–2408.
- AUCLERT, A., R. RIGATO, M. ROGNLIE, AND L. STRAUB (2024a): “New Pricing Models, Same Old Phillips Curves?” *The Quarterly Journal of Economics*, 139, 121–186.

- AUCLERT, A., M. ROGNLIE, AND L. STRAUB (2020): “Micro jumps, macro humps: Monetary policy and business cycles in an estimated HANK model,” *NBER Working Paper*.
- (2024b): “The intertemporal keynesian cross,” *Journal of Political Economy*, 132, 4068–4121.
- AUTOR, D., A. DUBE, AND A. MCGREW (2023): “The unexpected compression: Competition at work in the low wage labor market,” *NBER Working Paper*.
- BACH, G. L. AND J. B. STEPHENSON (1974): “Inflation and the Redistribution of Wealth,” *The Review of Economics and Statistics*, 1–13.
- BAILEY, M. J. (1956): “The welfare cost of inflationary finance,” *Journal of Political Economy*, 64, 93–110.
- BARDÓCZY, B. AND J. GUERREIRO (2023): “Unemployment Insurance in Macroeconomic Stabilization with Imperfect Expectations,” *UCLA Working paper*.
- BINETTI, A., F. NUZZI, AND S. STANTCHEVA (2024): “People’s Understanding of Inflation,” *NBER Working Paper*.
- BLANCHARD, O. AND J. GALÍ (2010): “Labor markets and monetary policy: A new keynesian model with unemployment,” *American Economic Journal: Macroeconomics*, 2, 1–30.
- BLANCO, A. AND A. DRENİK (2023): “How Does Inflation “Grease the Wheels” in a Frictional Labor Market?” *NBER Working Paper*.
- BLANCO, A., A. DRENİK, AND E. ZARATIEGUI (2024): “Nominal Devaluations, Inflation and Inequality,” *NBER Working Paper*.
- BURSTEIN, A. AND C. HELLWIG (2008): “Welfare costs of inflation in a menu cost model,” *American Economic Review*, 98, 438–443.
- CARATELLI, D. AND B. HALPERIN (2023): “Optimal monetary policy under menu costs,” *mimeo*.
- CHODOROW-REICH, G. AND L. KARABARBOUNIS (2016): “The cyclicity of the opportunity cost of employment,” *Journal of Political Economy*, 124, 1563–1618.
- CHODOROW-REICH, G., L. KARABARBOUNIS, AND R. KEKRE (2023): “The macroeconomics of the Greek depression,” *American Economic Review*, 113, 2411–2457.
- CHRISTIANO, L. J., M. S. EICHENBAUM, AND M. TRABANDT (2016): “Unemployment and business cycles,” *Econometrica*, 84, 1523–1569.
- CLARKE, F. H. (1990): *Optimization and nonsmooth analysis*, SIAM.
- COSTAIN, J., A. NAKOV, AND B. PETIT (2019): “Monetary policy implications of state-dependent prices and wages,” *Banco de Espana Working Paper*.

- DEL CANTO, F. N., J. R. GRIGSBY, E. QIAN, AND C. WALSH (2023): “Are Inflationary Shocks Regressive? A Feasible Set Approach,” *NBER Working Paper*.
- DOEPKE, M. AND M. SCHNEIDER (2006): “Inflation and the redistribution of nominal wealth,” *Journal of Political Economy*, 114, 1069–1097.
- ERCEG, C. J., D. W. HENDERSON, AND A. T. LEVIN (2000): “Optimal Monetary Policy with Staggered Wage and Price Contracts,” *Journal of monetary Economics*, 46, 281–313.
- FABERMAN, R. J., A. I. MUELLER, A. ŞAHIN, AND G. TOPA (2022): “Job search behavior among the employed and non-employed,” *Econometrica*, 90, 1743–1779.
- FELDSTEIN, M., J. GREEN, AND E. SHESHINSKI (1978): “Inflation and taxes in a growing economy with debt and equity finance,” *Journal of Political Economy*, 86, S53–S70.
- FERREIRA, C., J. M. LEIVA, G. NUÑO, Á. ORTIZ, T. RODRIGO, AND S. VAZQUEZ (2023): “The Heterogeneous Impact of Inflation on Households’ Balance Sheets,” *CESifo Working Paper*.
- FISCHER, S. AND F. MODIGLIANI (1978): “Towards an understanding of the real effects and costs of inflation,” *Review of World Economics*, 114, 810–833.
- FLOOD, S., M. KING, R. RODGERS, S. RUGGLES, J. R. WARREN, D. BACKMAN, A. CHEN, G. COOPER, S. RICHARDS, M. SCHOUWEILER, AND M. WESTBERRY (2023): “IPUMS CPS: Version 11.0 [Dataset],” Dataset, Minneapolis, MN: IPUMS.
- FRIEDMAN, M. (1969): *The optimum quantity of money: and other essays*, Aldine De Gruyter.
- (1977): “Nobel lecture: inflation and unemployment,” *Journal of political economy*, 85, 451–472.
- GALÍ, J. (2011): “The return of the wage Phillips curve,” *Journal of the European Economic Association*, 9, 436–461.
- GALÍ, J., F. SMETS, AND R. WOUTERS (2012): “Unemployment in an estimated New Keynesian model,” *NBER macroeconomics annual*, 26, 329–360.
- GERTLER, M., C. HUCKFELDT, AND A. TRIGARI (2020): “Unemployment fluctuations, match quality, and the wage cyclicality of new hires,” *The Review of Economic Studies*, 87, 1876–1914.
- GERTLER, M. AND J. LEAHY (2008): “A Phillips curve with an Ss foundation,” *Journal of Political Economy*, 116, 533–572.
- GERTLER, M., L. SALA, AND A. TRIGARI (2008): “An estimated monetary DSGE model with unemployment and staggered nominal wage bargaining,” *Journal of Money, Credit and Banking*, 40, 1713–1764.

- GERTLER, M. AND A. TRIGARI (2009): “Unemployment fluctuations with staggered Nash wage bargaining,” *Journal of political Economy*, 117, 38–86.
- GOLOSOV, M. AND R. E. LUCAS (2007): “Menu costs and Phillips curves,” *Journal of Political Economy*, 115, 171–199.
- GROSSMAN, S. J. AND O. D. HART (1986): “The costs and benefits of ownership: A theory of vertical and lateral integration,” *Journal of political economy*, 94, 691–719.
- HAJDINI, I., E. S. KNOTEK, J. LEER, M. PEDEMONTE, R. W. RICH, AND R. SCHOENLE (2023): “Low passthrough from inflation expectations to income growth expectations: why people dislike inflation,” *FRB of Cleveland Working Paper*.
- HART, O. AND J. MOORE (1990): “Property Rights and the Nature of the Firm,” *Journal of political economy*, 98, 1119–1158.
- HAZELL, J. AND B. TASKA (2024): “Downward rigidity in the wage for new hires,” *American Economic Review*.
- İMROHOROĞLU, A. (1992): “The welfare cost of inflation under imperfect insurance,” *Journal of Economic Dynamics and Control*, 16, 79–91.
- JACK, B. K., K. McDERMOTT, AND A. SAUTMANN (2022): “Multiple Price Lists for Willingness to Pay Elicitation,” *Journal of Development Economics*, 159, 102977.
- JO, Y. J. (2019): “Downward nominal wage rigidity in the United States,” *Job Market Paper*.
- KESSEL, R. A. AND A. A. ALCHIAN (1960): “The meaning and validity of the inflation-induced lag of wages behind prices,” *The American Economic Review*, 50, 43–66.
- LORENZONI, G. AND I. WERNING (2023a): “Inflation is conflict,” *NBER Working Paper*.
- (2023b): “Wage price spirals,” *Brookings Papers in Economic Activity*.
- LUCAS, R. (2000): “Inflation and welfare,” *Econometrica*, 68, 247–74.
- MANKIW, G. (2020): *Principles of Macroeconomics*, Cengage Learning.
- MILGROM, P. AND I. SEGAL (2002): “Envelope theorems for arbitrary choice sets,” *Econometrica*, 70, 583–601.
- NAKAMURA, E. AND J. STEINSSON (2010): “Monetary Non-Neutrality in a Multisector Menu Cost Model,” *The Quarterly journal of economics*, 125, 961–1013.
- NAKAMURA, E., J. STEINSSON, P. SUN, AND D. VILLAR (2018): “The Elusive Costs of Inflation: Price Dispersion During the US Great Inflation,” *The Quarterly Journal of Economics*, 133, 1933–1980.

- PALLOTTI, F., G. PAZ-PARDO, J. SLACALEK, O. TRISTANI, AND G. L. VIOLANTE (2023): “Who Bears the Costs of Inflation? Euro Area Households and the 2021–2022 Shock,” *NBER Working Paper*.
- PEW RESEARCH CENTER (2022): “By a Wide Margin, Americans View Inflation as the Top Problem Facing the Country Today,” .
- (2023): “Inflation, Health Costs, Partisan Cooperation Among the Nation’s Top Problems,” .
- PILOSSOPH, L., J. RYNGAERT, AND J. WEDEWER (2024): “The search costs of inflation,” *mimeo*.
- PILOSSOPH, L. AND J. M. RYNGAERT (2024): “Job Search, wages, and inflation,” .
- SHILLER, R. J. (1997): “Why Do People Dislike Inflation?” in *Reducing inflation: Motivation and strategy*, University of Chicago Press, 13–70.
- SHIMER, R. (2005): “The cyclical behavior of equilibrium unemployment and vacancies,” *American economic review*, 95, 25–49.
- SILVA, J. I. AND M. TOLEDO (2009): “Labor turnover costs and the cyclical behavior of vacancies and unemployment,” *Macroeconomic Dynamics*, 13, 76–96.
- SMETS, F. AND R. WOUTERS (2007): “Shocks and frictions in US business cycles: A Bayesian DSGE approach,” *American economic review*, 97, 586–606.
- STANTCHEVA, S. (2023): “How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible,” *Annual Review of Economics*, 15, 205–234.
- (2024): “Why Do We Dislike Inflation?” *Brookings Papers on Economic Activity*.
- TAKAHASHI, S. (2018): “State Dependency in Price and Wage Setting,” *47th issue (February 2017) of the International Journal of Central Banking*.